SCALABLE INTERNET TRAFFIC FILTERING PLATFORM

Centre for Internet Studies & Artificial Intelligence

Feature list, Datasheet and User manual



SPONSORING AGENCY: NSCS

IMPLEMENTATION AGENCY:

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1 INTRODUCTION

1.1 PURPOSE

The Internet Traffic Filtering Solution, developed by Centre for Internet Studies and Artificial Intelligence, Amrita Vishwa Vidyapeetham, to provide regulatory bodies with the ability to effectively manage and control internet traffic at the gateways of Internet Service Providers (ISPs). It does so by adhering to specified control policies which dictate the traffic that can pass through these gateways. The highly scalable Traffic Filtering appliance has a throughput up to 400G (200G inbound and 200G outbound).

An appliance capable of 800G is under testing. Upon successful completion of these tests, assuring the appliance's robust performance and reliability, it will be officially introduced to our product line. This enhanced version promises to significantly augment the capacity to manage and filter higher volumes of internet traffic.

2 OVERALL DESCRIPTION

2.1 ENVIRONMENTAL SPECIFICATIONS

2.1.1 Operating temperature: 10°C to 35°C
2.1.2 Operating humidity: 8-80% (non-condensing)
2.1.3 Non-operating temperature: -40°C to 60°C
2.1.4 Non-operating humidity: 20%-95% (non-condensing)

2.2 ELECTRICAL SPECIFICATIONS

2.2.1 2 x 1200W 1+1 redundant power supplies
2.2.2 Input voltage: 200-240V
2.2.3 Maximum current consumption: 7.08A
2.2.4 Operating frequency: 47-63Hz
2.2.5 Average power consumption: 550W
2.2.6 Peak power consumption: 1200W

2.3 PHYSICAL SPECIFICATIONS

2.3.1 W x H x D: 438 x 43.5 x 710 mm (Without optical bypass unit)
2.3.2 W x H x D: 438 x 87 x 710 mm (With optical bypass unit)
2.3.3 Rack units: 1U (Without optical bypass unit)
2.3.4 Rack units: 2U (With optical bypass unit)
2.3.5 Net Weight: 13 kg

2.4FILTERING SPECIFICATIONS

2.4.1 Total throughput capacity: 400Gbps
2.4.2 Blocklist capacity: 10 million IPs/Domains
2.4.3 Latency: <5ms

2.5REDUNDANCY

2.5.1 Hot pluggable dual power supply support.
2.5.2 Supports redundant cooling with hot pluggable units.
2.5.3 Supports up to 4 SSD/HDD with mirroring RAID support.
2.5.4 Supports two appliances to be in HA configuration provided ISP supports HA/load balance on source and destination side.
2.5.5 Configurable optical bypass switch with auto- bypass function.

2.6MTBF (MEAN TIME BEFORE FAILURE)

10 years (20 years with redundancy takeover)

2.7 AVERAGE LIFE OF EQUIPMENT

10-12 years

a. AVAILABILITY

99.998622%

b. I/O

Management ports: 2 x 10G RJ45

Available filter port configurations:

1. 4 x 100G
2. 8 x 25G
3. 4 x 25G + 2 x 100
4. 4 x 25G + 8 x 10G/1G (8 ports will be of same link speed)
5. 16 x 10G/1G (All 16 ports will be of same link speed)
6. 8 x 10G/1G (8 ports will be of same link speed) + 8 x 10G/1G (8 ports will be of same link speed)
7. 2 x 100G + 8 x 25G/10G/1G (8 ports will be of same link speed)

3 SYSTEM FEATURES

3.1 USER INTERFACE

3.1.1 Dashboard

The dashboard features live statistics.

- 3.1.1.1 Total ingress bandwidth live graph
- 3.1.1.2 Total egress bandwidth live graph
- 3.1.1.3 Total ingress packet live graph
- 3.1.1.4 Total egress packet live graph
- 3.1.1.5 Ingress bandwidth live meter
- 3.1.1.6 Egress bandwidth live meter
- 3.1.1.7 Live statistics of blocked packets
- 3.1.1.8 Live statistics of blocked bandwidth

3.1.2 Reports

The report page provides traffic summary, historic and live statistics of individual ports and rule hit statistics.

- 3.1.2.1 Traffic summary: The traffic summary includes the bandwidth and packet rate for each inbound and outbound port pair during a user-specified timeframe.
- 3.1.2.2 Live and historic port statistics: Real-time and historical data of bandwidth and packet rates for each individual port.
- 3.1.2.3 Blocked traffic: Detailed statistics for rule and feed hits during a user-selected time frame (Raw statistics is downloadable).

3.1.3 Access control

The Access Control interface grants the user comprehensive control over rules, domain and IP feed management, source networks, and commit history.

3.1.3.1 Rules: Rule management page allow the user to add blacklist (Block specified traffic) and whitelist (Allow only specified traffic) rules. These rules can be formed with a combination of IP list, domain list, app list, source network (The rule will be applied only to the specific source network), transport layer port and port/interface. Any added rules will only be activated upon user commitment.

3.1.3.2 Object management: Object management allows the user to manage the source network and IP/Domain feed management. The feed management page lets the user to add IP, subnet, FQDN, wildcard domain, ports, application (Including application category). User can add the IP/subnet of the sources which is to be applied to a rule in the source management page. The user has the option to upload bulk data using file upload as well.

3.1.3.3 Commit history: The commit history page provides a full overview of all commits, including the additions and deletions of objects and rules done in the commit.

3.1.4 Administration

The administration page allows the user to manage the users, central console server, log servers and applications.

3.1.4.1 User management: The User Management portal allows for the addition, deletion, and modification of users. The administrator has the discretion to determine user privileges and to grant or revoke these as necessary. Additionally, the page provides access to login logs for a specified timeframe.

3.1.4.2 Application signature management: This feature permits the user to add the signatures of new applications manually in offline mode by uploading a file that contains the application signatures.

3.1.4.3 Log Forwarding: Syslog servers can be configured here to facilitate the forwarding of logs for storage and analysis purposes.

3.1.4.4 Central console management: Central Console Management enables users to add or remove central console servers. This central console serves act as a unified platform for managing multiple appliances from a singular location.

3.2 COMMIT

The commit feature allows the user to commit the changes to the data plane. This will help to prevent unintended alterations in rules from reflecting in the filtering appliance’s data plane.

3.3 OPTICAL BYPASSING

The Optical Bypassing Feature is designed to enhance system resilience and continuity during periods of signal loss. In such instances, the feature will facilitate optical bypass of the appliance. This bypass module is automated to initiate a bypass in scenarios such as power failure of the appliance, unintentional shutdowns, or can be manually invoked by the user using a dedicated bypass button.

To ensure the prevention of accidental bypass activation, the design requires the use of a pointed pin to press the bypass button.

Furthermore, the feature is programmed to continuously monitor signal strength. If signal strength descends below a predetermined threshold, the system will automatically initiate the bypass. This maintains system functionality and prevents service disruption even under less-than-optimal signal conditions.

The Optical Bypassing Feature is designed to engage within a matter of nanoseconds, thereby preserving session continuity and preventing potential data loss.

3.4 TRANSPARENT MODE

The transparent mode allows the appliance to operate undetected, providing filtering without needing an IP address.

The filtering ports in the appliance doesn't require an IP address, allowing it to invisibly pass through data while still filtering.

The transparent mode ensures that adding the appliance to an existing network can be done without redesigning the IP addressing scheme.

3.5 BLACKLIST/WHITELIST

The traffic filtering platform is capable of Blacklisting/Whitelisting with the following objects:

3.5.1 FQDN

3.5.2 Domain Wildcard

3.5.3 IPv4

3.5.4 IPv4 Subnets via prefix

3.5.5 IPv6

3.5.6 IPv6 Subnets via prefix

3.5.7 Applications

3.5.8 Ports

3.6 PROPOSED DEPLOYMENT ARCHITECTURE

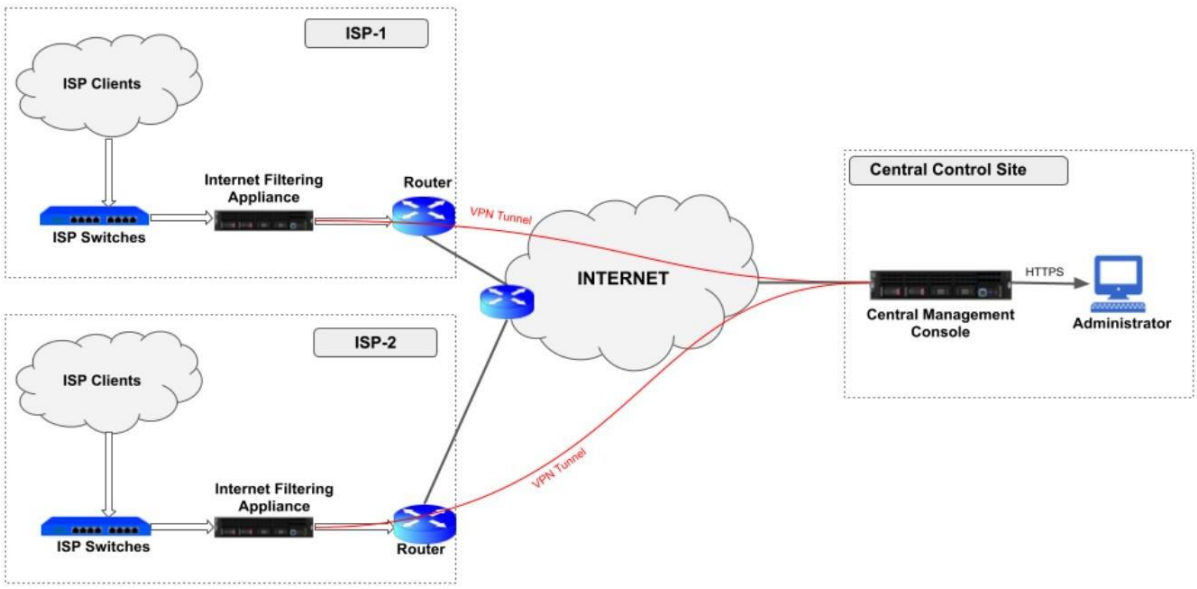


Figure 3.1

1. One or multiple appliances can be deployed at the ISP Client sites, between the client devices and the internet gateway.
2. Each of the the appliances can be connected to a single central console site.
3. The central console site can monitor and manage individual appliances at each ISP sites.
4. Traffic between the central console and the filtering appliance is protected by TLS encryption, further reinforced by the use of an additional VPN tunnel to ensures security.
5. Each appliance's central console management connection links to the destination side, ensuring uninterrupted access to the central console, even if internet blocking via firewall occurs.
6. In scenarios where the appliance's central console management connection cannot be established on the destination side and is limited to the source side, the whitelist feature can be utilised to maintain connectivity. This connection is safeguarded through Asymmetric key encryption within the VPN, providing robust security regardless of the connection's origin point.

4. Getting Started

This section explains how to get started with CISAI Internet Filtering Solution.

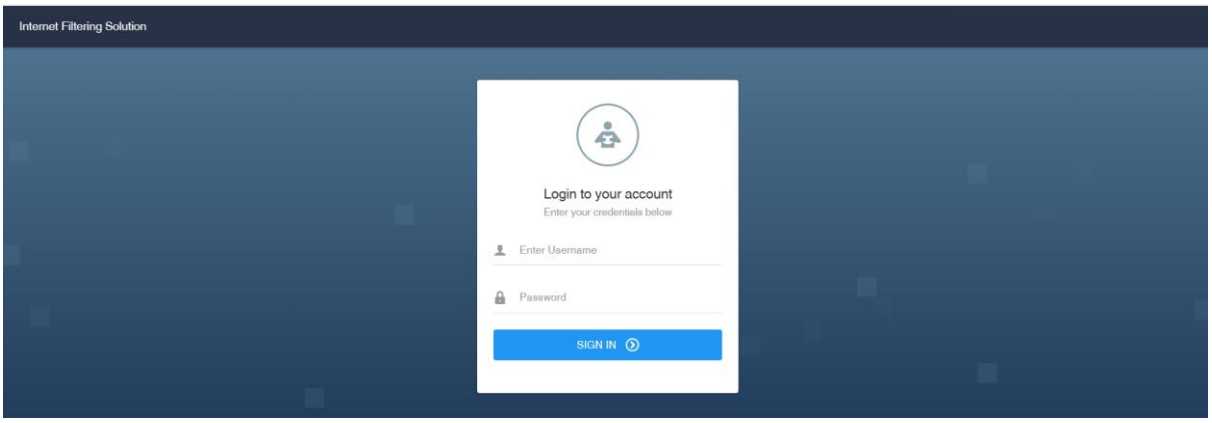


Figure 4-1

4.1 To access the portal

a. Enter the username

b. Enter the password

c. Click SIGN IN button.

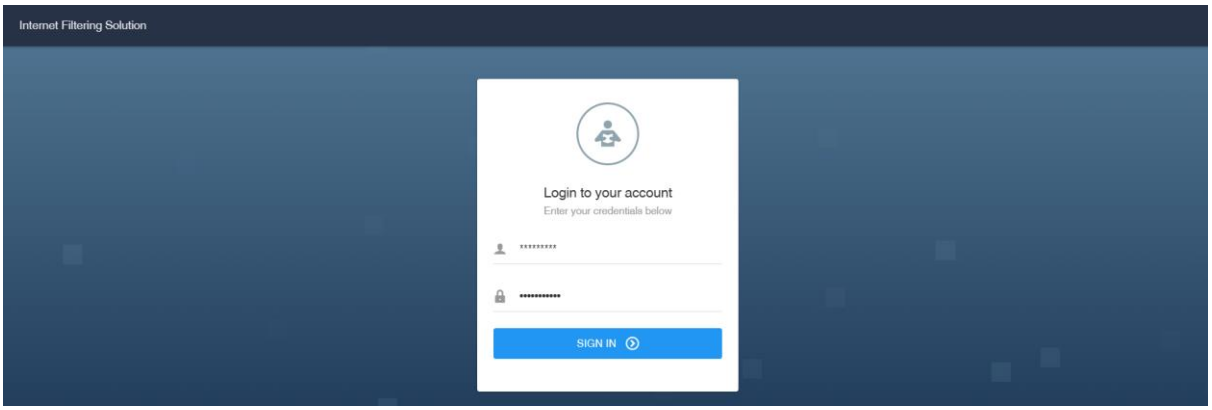


Figure 4-2

On clicking the SIGN IN button, they are redirected to CISAI Internet Filtering Solution GUI

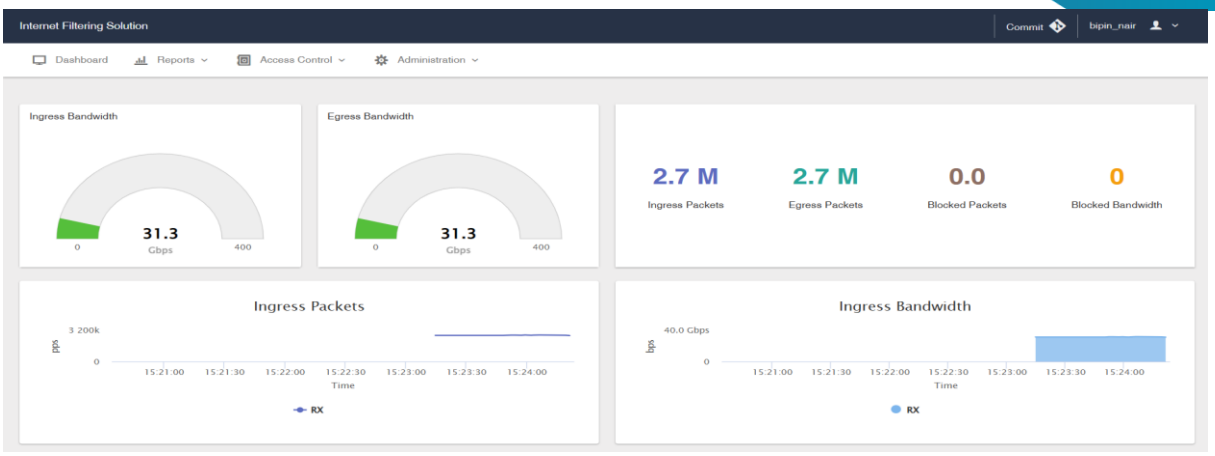


Figure 1-3

The GUI has the following menus which provides access to configuration options and statistical visualization:

a. **Dashboard** - The dashboard displays various widgets that display important system information. For more information, kindly refer Chapter 2 of this user manual.

b. **Reports** - The reports menu provides consolidated statistics on network traffic, allowed and blocked traffic, interface status and providing live statistics and historic statistics of the various ports monitored by the internet filtering solution

c. **Access Control** - It provides ability to the user to add source device, add applications whose traffic are to be filtered and creating filtering rules. The user can also monitor the changes made to the filtering rules.

d. **Administration** - The user(s) with administrator privilege can add or remove users, change their privileges or roles, enable or disable log forwarding via syslog servers, and central console management.

Filters are used to locate a specific set of information or content in a network packets. They can be particularly useful for obtaining traffic statistics through specific port pair combination. The filtering options vary, depending on the type of statistics required. The administrator can apply multiple filters at the same time.

4.2 To manually create filter

a. Select the field to filter by

b. Click the drop-down list in front of the filter

c. Enter the value to filter by, adding modifiers as needed

d. Click submit to apply the filter

4.3 Entering values

Numerous fields in the GUI require text strings or numbers to be entered when configuring the rules. When entering values in the GUI, you will be prevented from entering invalid characters, and a warning message will be shown explaining what values are not allowed. If invalid values are entered, the setting will be rejected when you apply it.

4.3.1 Text Strings

Text strings are used to name entities in creating source networks, feed management and rules. The tool used UTF-8 character encoding. Encodings usually vary based on the client's operating system or input language.

The following characters cannot be used in text strings, as they present cross-site scripting (XSS) vulnerabilities:

" - double quotes

' - single quote

> - greater than

< - less than

The GUI text fields prevent XSS vulnerable characters from being added.

4.3.2Numbers

Numbers are used to set sizes, rated, addresses, port numbers, priorities, and other such numeric values. They can be entered as a series of digits (without commas or spaces), in a dotted decimal format (such as IP addresses), or separated by colons (such as MAC addresses). Most numeric values use base 10 numbers, while some use hexadecimal values.

The GUI prevents invalid numbers from being entered.

4.3.3 Permissions

Administrator or access profiles control what menus and widgets an administrator can access by assigning read, write, or no access to each area.

Read access is required to view configurations. Write access is required to make configuration changes. Depending on your account's profile, you may not have access to all functionalities of this tool.

To have access to all menus and widgets, an administrator account with the super_admin profile must be used, such as the admin account. Accounts assigned the super_admin profile are similar to the root administrator account. They have full permission to view and change all configuration options, including viewing and changing other administrator accounts.

To increase account security, set strong passwords for all administrator accounts, and change the passwords regularly.

4.4.4 To obtain more information about CISAI Internet Filtering Solution

a. Click the drop-down arrow at the top-right of the page.



Figure 4-4

b. Click About

c. To change password

d. Click the drop-down arrow at the top-right of the page.

e. Click Password

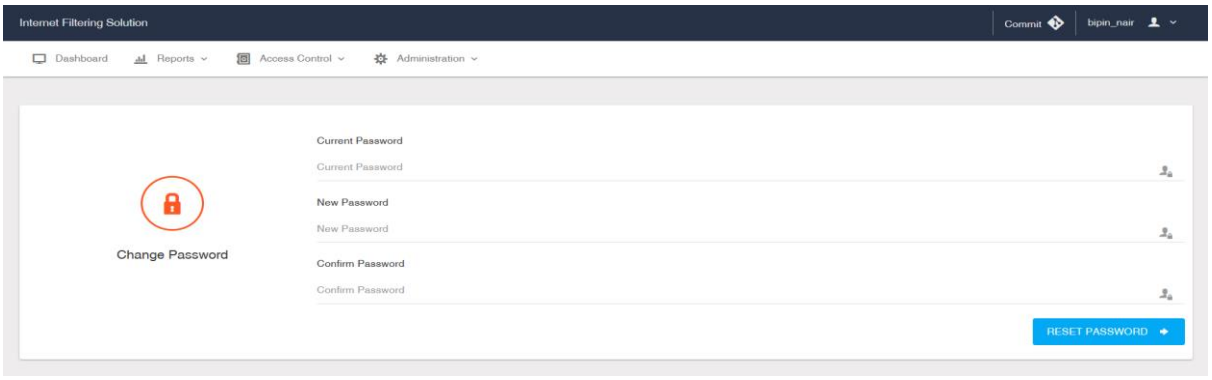


Figure 4-5

f. Enter the current password.

g. Enter the new password

h. Re-enter the new password

i. Click RESET PASSWORD button

4.4.5 To logout from Internet Filtering Solution

a. Click the drop-down arrow at the top-right of the page.

b. Click Logout

Note: If the user does not perform any interaction with CISAI Internet Filtering Solution for 5 to 10 minutes, then they are automatically logged out. The user will be prompted to log in once again. When the user logs in, their session starts at the same page where they left last.

5. Dashboard

The window that is presented to the user when they log into the tool is the Dashboard.

To access dashboard from other tabs, the user can click the Dashboard menu in the menu bar.

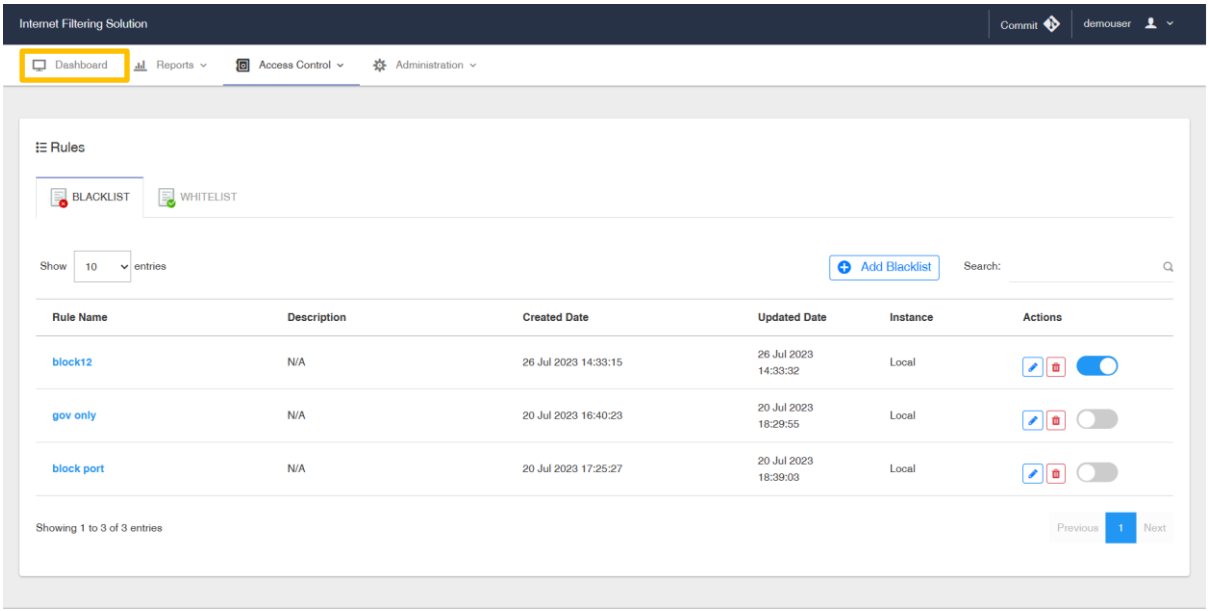


Figure 5-1

5.1. Ingress Bandwidth

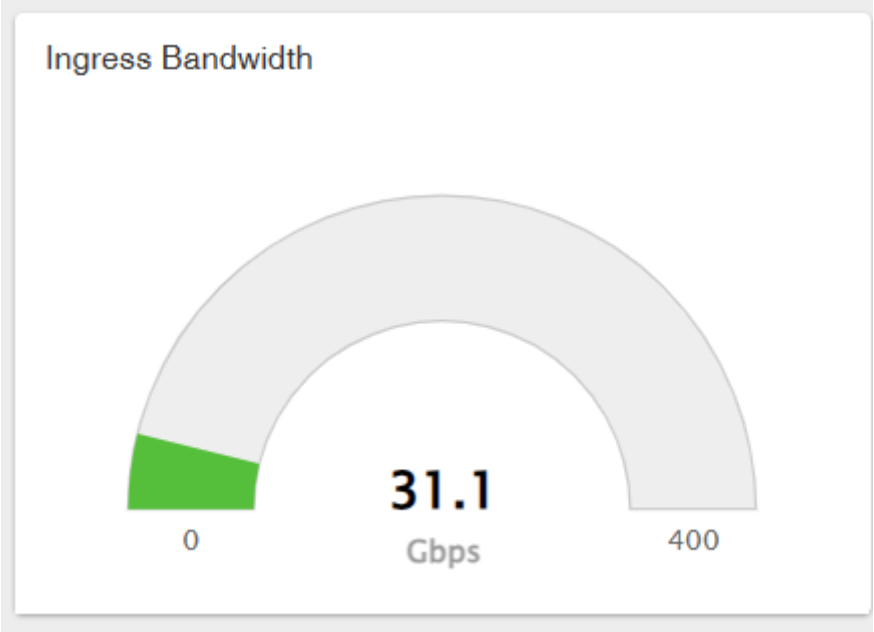


Figure 5-2

The user can obtain data for the download speed in this widget . The value represented is in Gigabits per second (Gbps).

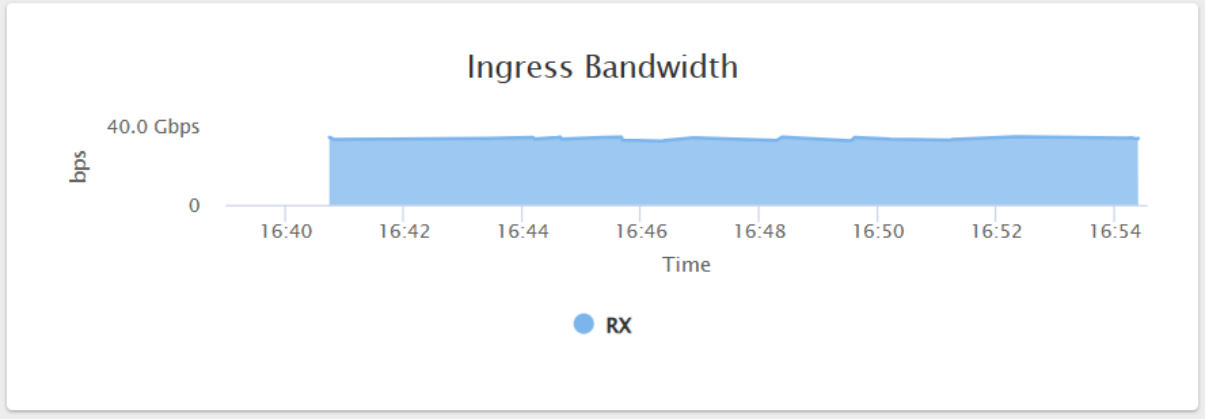


Figure 5-3

The user can also view the timeline graph of ingress bandwidth for a given period of time in the above widget.

5.2 Egress Bandwidth

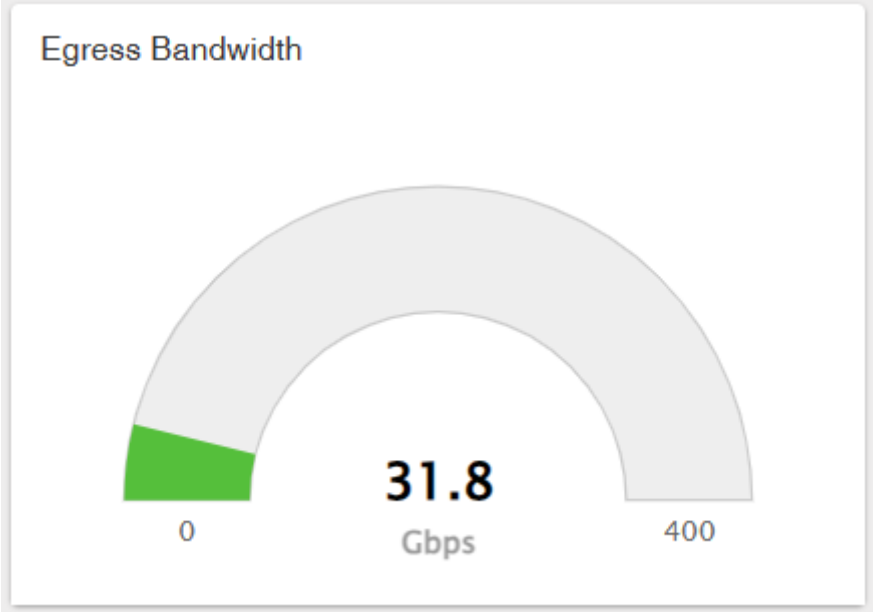


Figure 5-4

The user can obtain data for the upload speed in the widget.

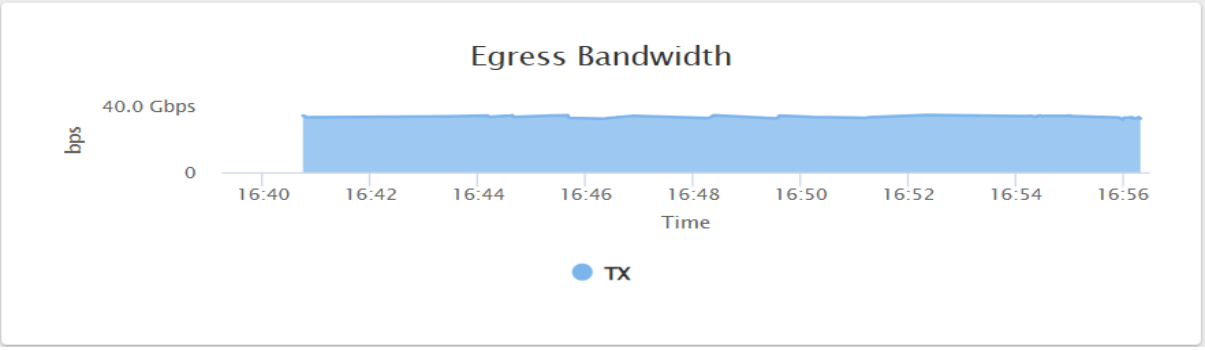


Figure 5-5

The user can also view the timeline graph of egress bandwidth for a given period of time in the above widget

5.3 Ingress Packets

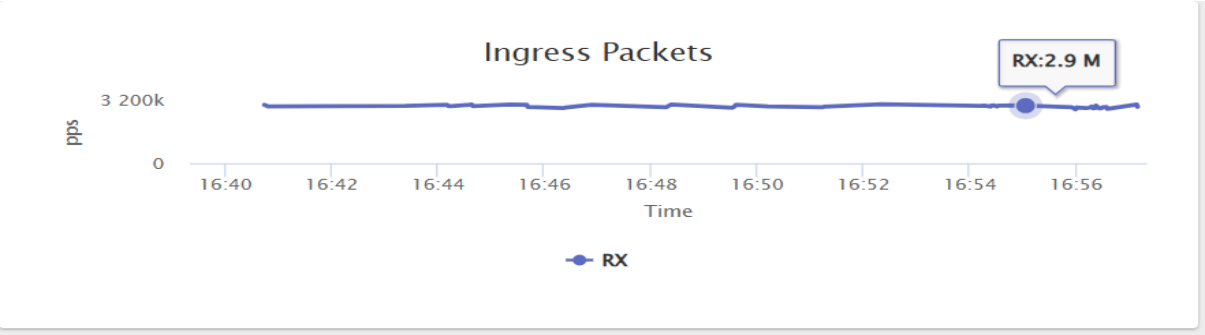


Figure 5-6

The user can obtain timeline graph of the packets of data received at a given period of time in this widget.

5.4 Egress Packets

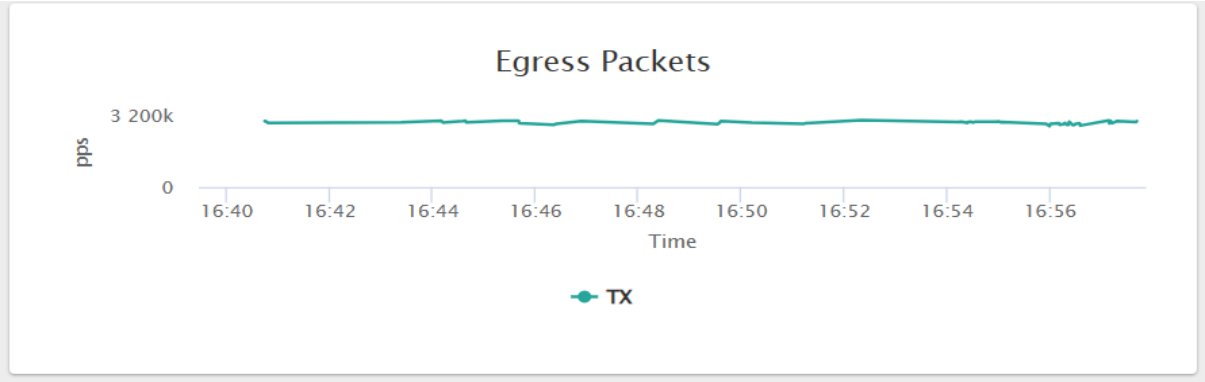


Figure 5-7

The user can obtain timeline graph of packets of data transmitted at a given period of time in this widget.

5.5 Consolidated packet view



Figure 5-8

In t his widget, the user can obtain a consolidated view of the count of ingress packets, egress packets, blocked packet and blocked bandwidth.

6. Reports

When the user clicks the Reports menu, the following submenus are presented on the screen.

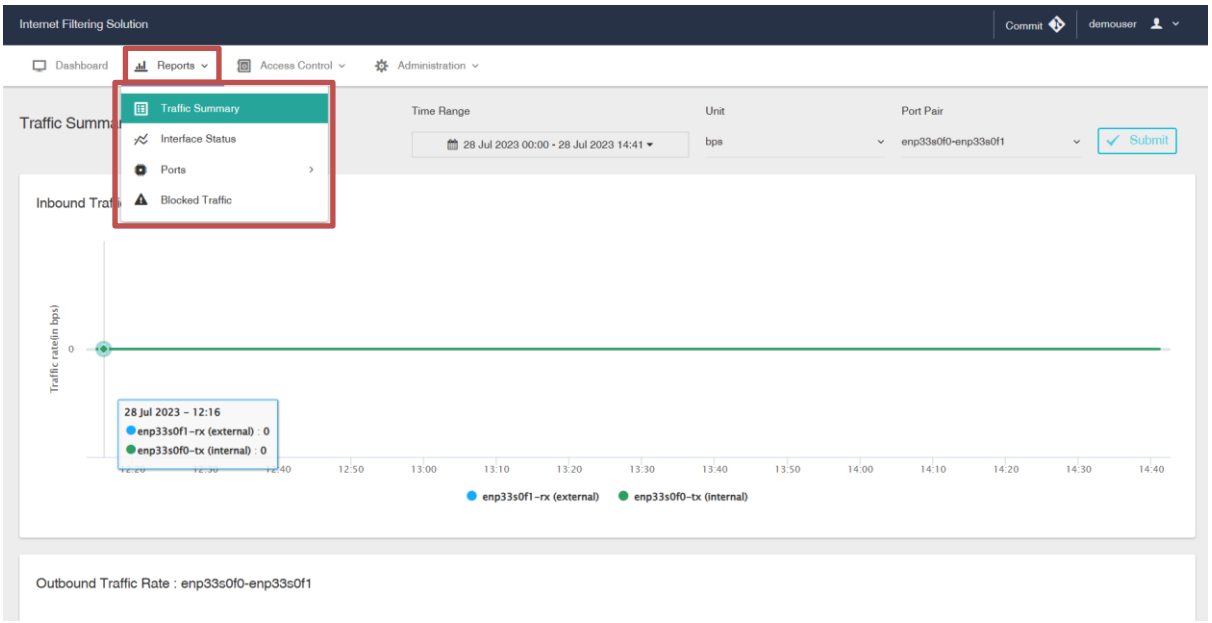


Figure 6-1

6.1 To view Traffic Summary

a. Click Reports menu

b. Click Traffic Summary submenu

c. In the traffic summary page, the following filter are available at the top right portion

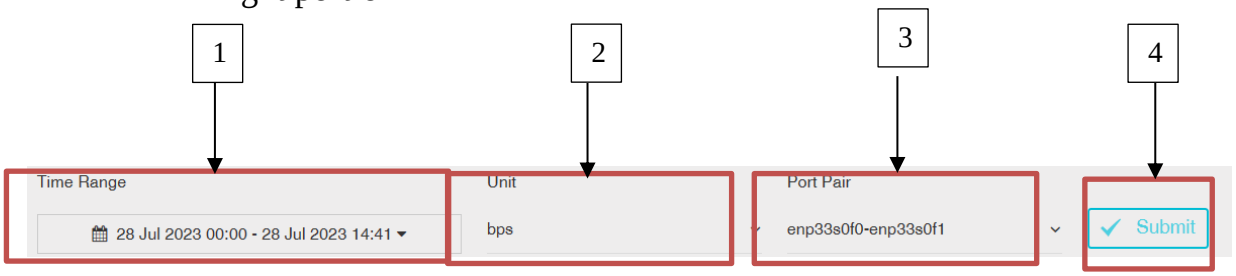


Figure 6-2

The user can set the date and time range for which the traffic summary is to be obtained. When the user clicks the Time Range filter, the following drop down appears.

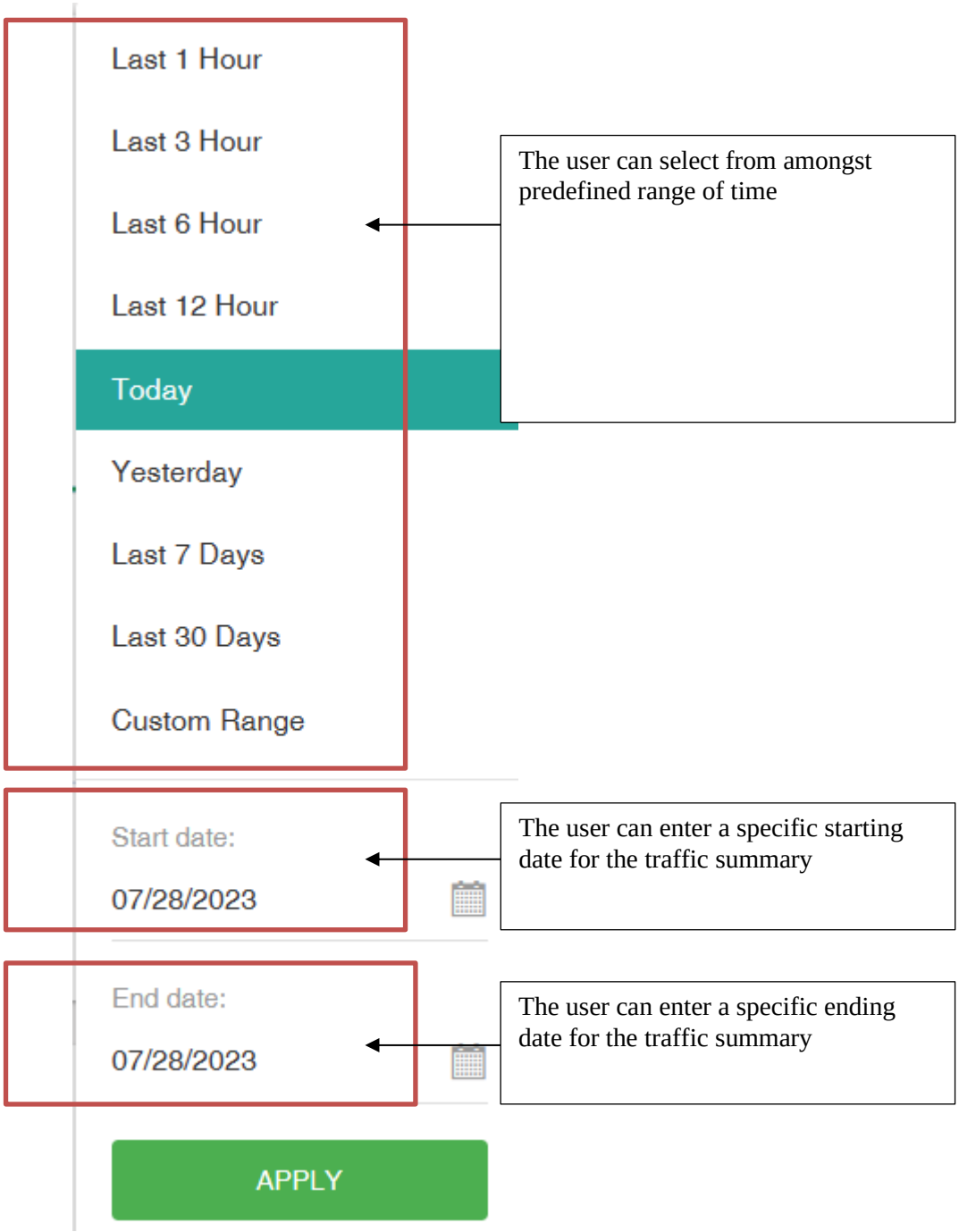


Figure 6-3

The user can set the unit as either bps(bytes per second) or pps(packets per second)

The user can select the port pair for which traffic summary is to be obtained.

When the user clicks the Apply button, the filter conditions are executed and corresponding traffic summary is presented to the user.

In the Traffic Summary page, the following widgets are available:

6.2 Inbound Traffic Rate

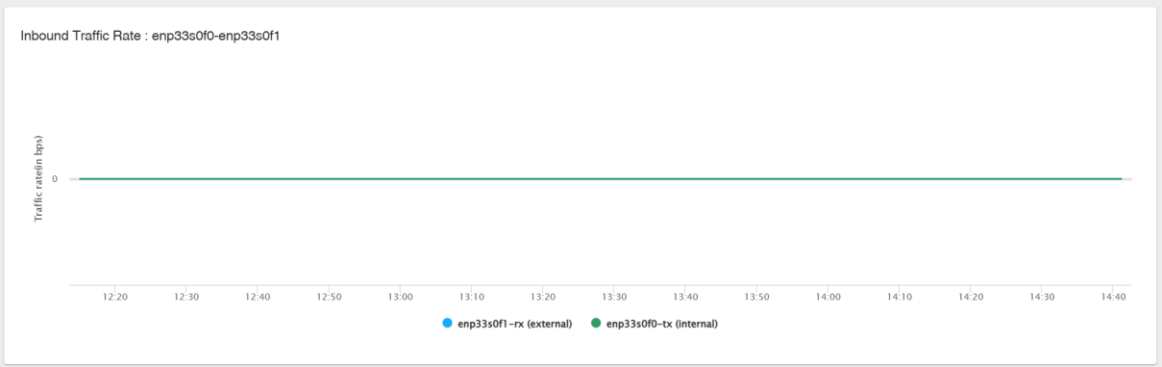


Figure 6-4

This widget provides the timeline graph of inbound traffic flow (in terms of the units selected) for the selected port-pair network and date and time range.

Note: Inbound traffic implies traffic entering into the network where the Internet Filtering Solution is deployed.

Note: The legends in the graph are interactive. The user can enable / disable the entries in the legend by clicking on the corresponding entry. If any entry is disabled, its displayed in light grey text. If its enabled, then the text is displayed in coloured icon and bold text.

6.3 Outbound Traffic Rate

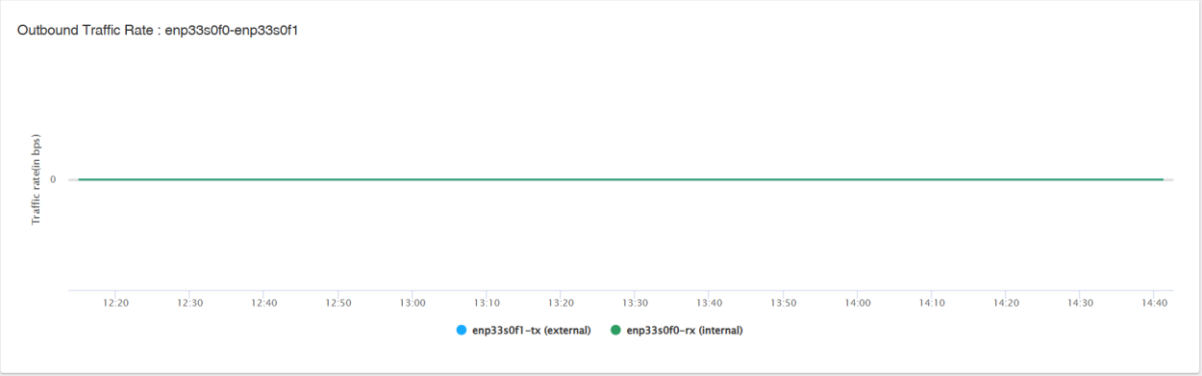


Figure 6-5

This widget provides the timeline graph of inbound traffic flow (in terms of the units selected) for the selected port-pair network and date and time range.

Note: Outbound traffic implies traffic exiting from the Internet Filtering Solution.

Note: The legends in the graph are interactive. The user can enable / disable the entries in the legend by clicking on the corresponding entry. If any entry is disabled, its displayed in light grey text. If its enabled, then the text is displayed in coloured icon and bold text.

Note: When the user hovers their mouse pointer over any point in the graph, the traffic rate for the corresponding time period is shown on the graph



Figure 6-6

6.4 Interface status

The following filters are provided at the top right portion of the Interface Status Page

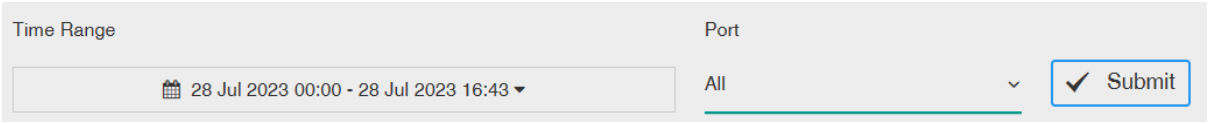


Figure 6-7

6.5 Live Interface Status

The user can view the current status of the ports in this widget. If the corresponding port is active, it is displayed in green colour and if it is inactive then it is displayed in red colour.



Figure 6-8

Note: The interfaces are automatically populated when connected to the internet filtering solution. Based on the ping, the status of the interface is identified.

6.6 Interface Status Timeline



Figure 6-9

The user can obtain timeline graph showing the behavior of the ports during the selected time period. This graph shows square wave form only when there is any change in the behavior of the interface (i.e. On or OFF).

The user can obtain interface status on hovering the mouse at the specific point on the graph.

6.7 Interface Status Table

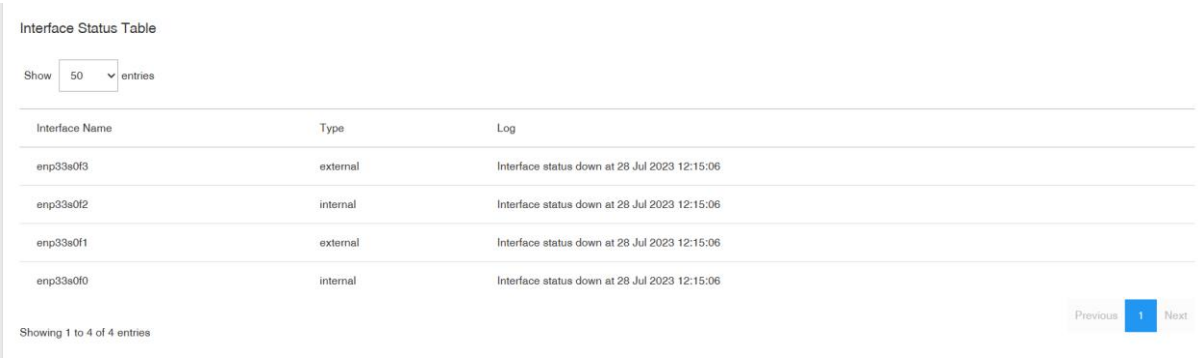


Figure 6-10

The user can obtain brief details about the interface in tabular form in this widget. It provides log report about the status of the interface as well as record of all changes that happened to each interface.

6.8 Ports

There are two options under Ports submenu in the Reports menu

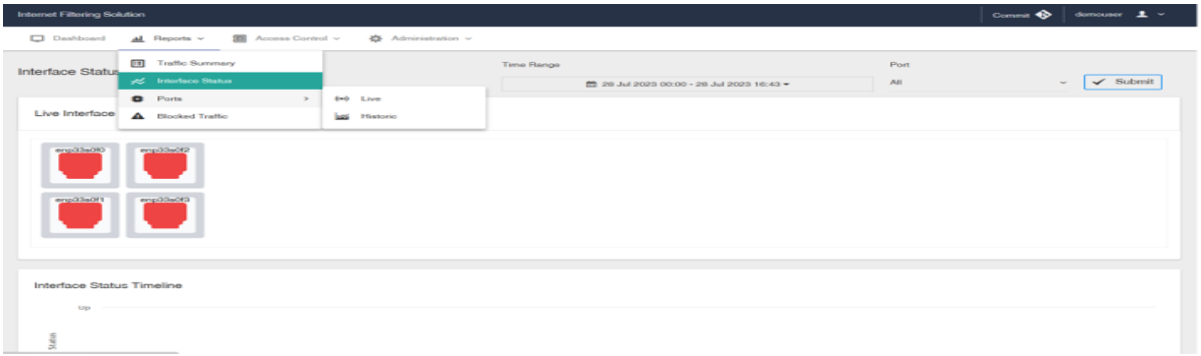


Figure 6-11

6.9 Live port statistics

The user can obtain live traffic statistics in this page.

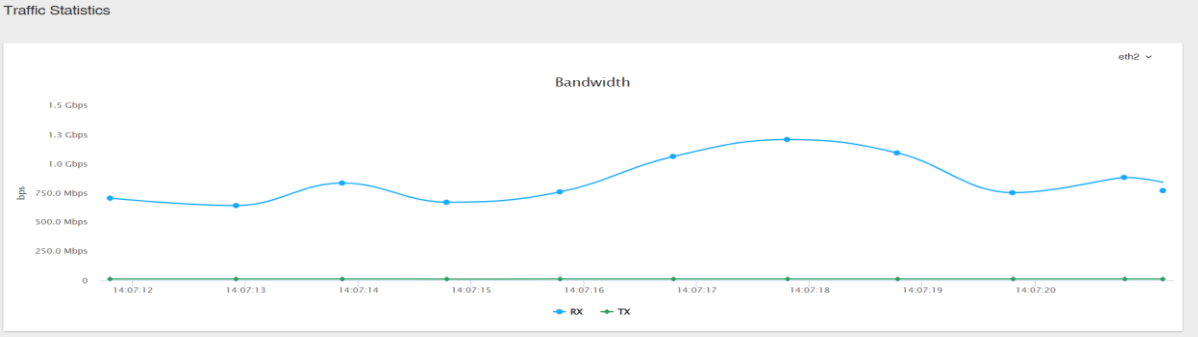


Figure 6-12

The user can select from the list of ports that can be obtained by clicking on the port name on the top-right of the corresponding visualization

The visualization provides statistics of bandwidth and packets.

In the timeline graph, RX represents ingress packets and TX represents egress packets

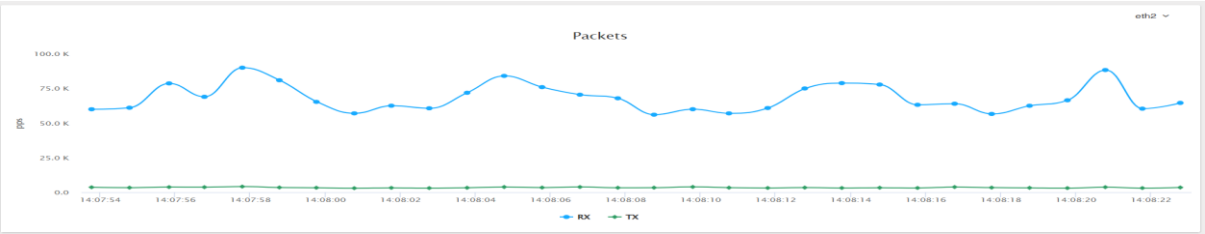


Figure 6-13

6.10 Historic port statistics

The user can obtain historic data (i.e. past statistics for ingress bandwidth, ingress packets, egress bandwidth and egress packets) by clicking this option.

6.11 Traffic Rate

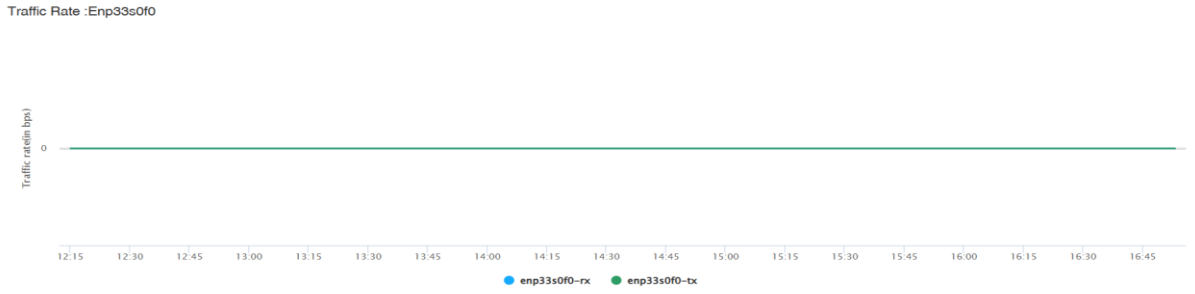


Figure 6-14

This widget provides a timeline graph of the traffic rate for ingress traffic and egress traffic for the given date and time range and port settings.

6.12 Traffic Stats

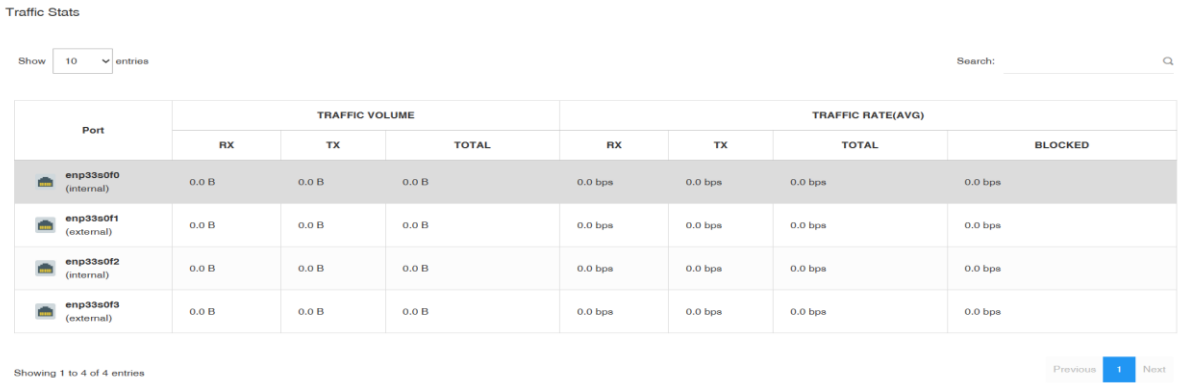


Figure 6-15

The user can obtain statistics of ingress and egress traffic volume and average ingress and egress traffic rate using this widget. The user can also obtain statistics on blocked bandwidth for the corresponding port in this widget.

6.13 Blocked Traffic

The user can obtain statistics about the blocked traffic for a selected date and time range and port using this option

The following widgets are available in the Blocked Traffic page

6.13.1 Total bytes

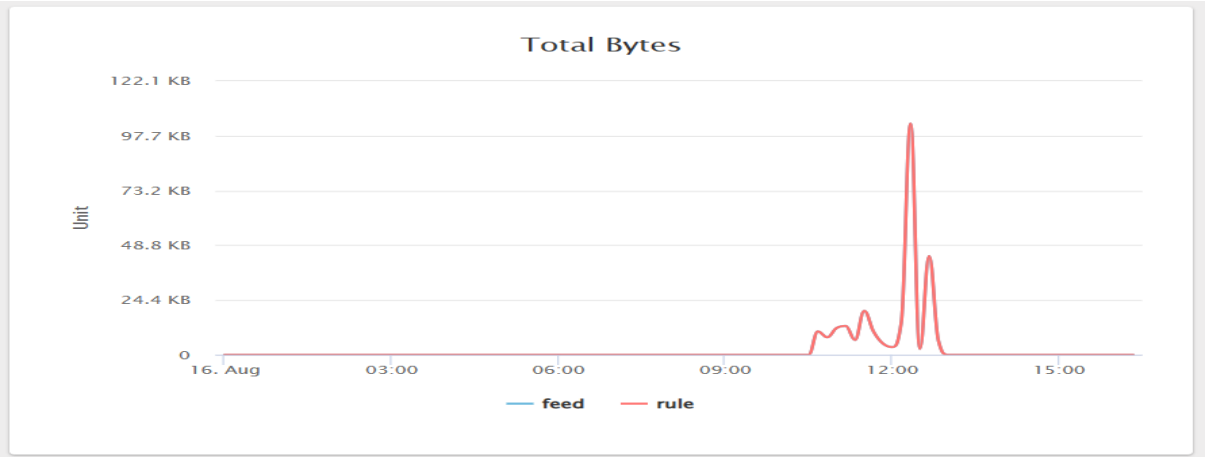


Figure 6-16

6.13.2 Total Packets

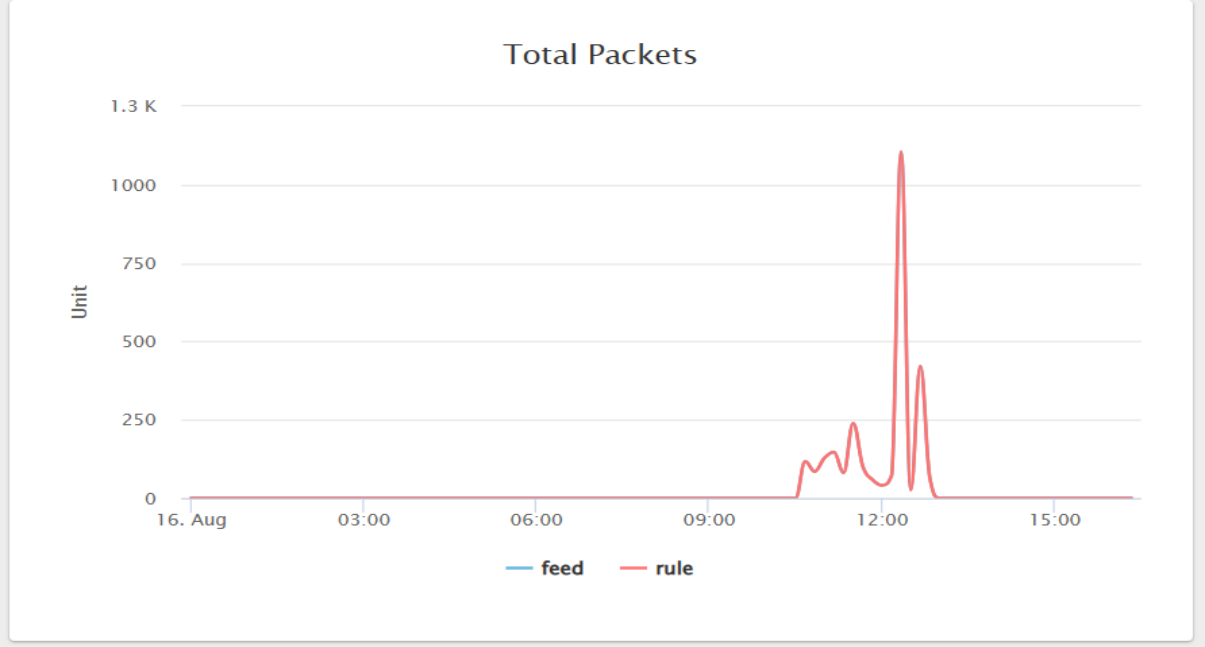


Figure 6-17

6.13.3 Rule Stats

Rule Stats

Search:

#	Rule Name	Bandwidth	Total Packets
1	amazontest	6.8 KB	633
2	facebooktest	7.0 KB	514
3	truecaller test	5.1 KB	372
4	jiosaavn	2.5 KB	248
5	chingari test	2.8 KB	240
6	inkitt test	1.7 KB	162
7	shopsy test	1.6 KB	147
8	whatsapp test	1.2 KB	117
9	bbc test	1.4 KB	109
10	bbctest	1.2 KB	96

Previous 1 2 Next

Figure 6-18

6.13.4 Feed Stats

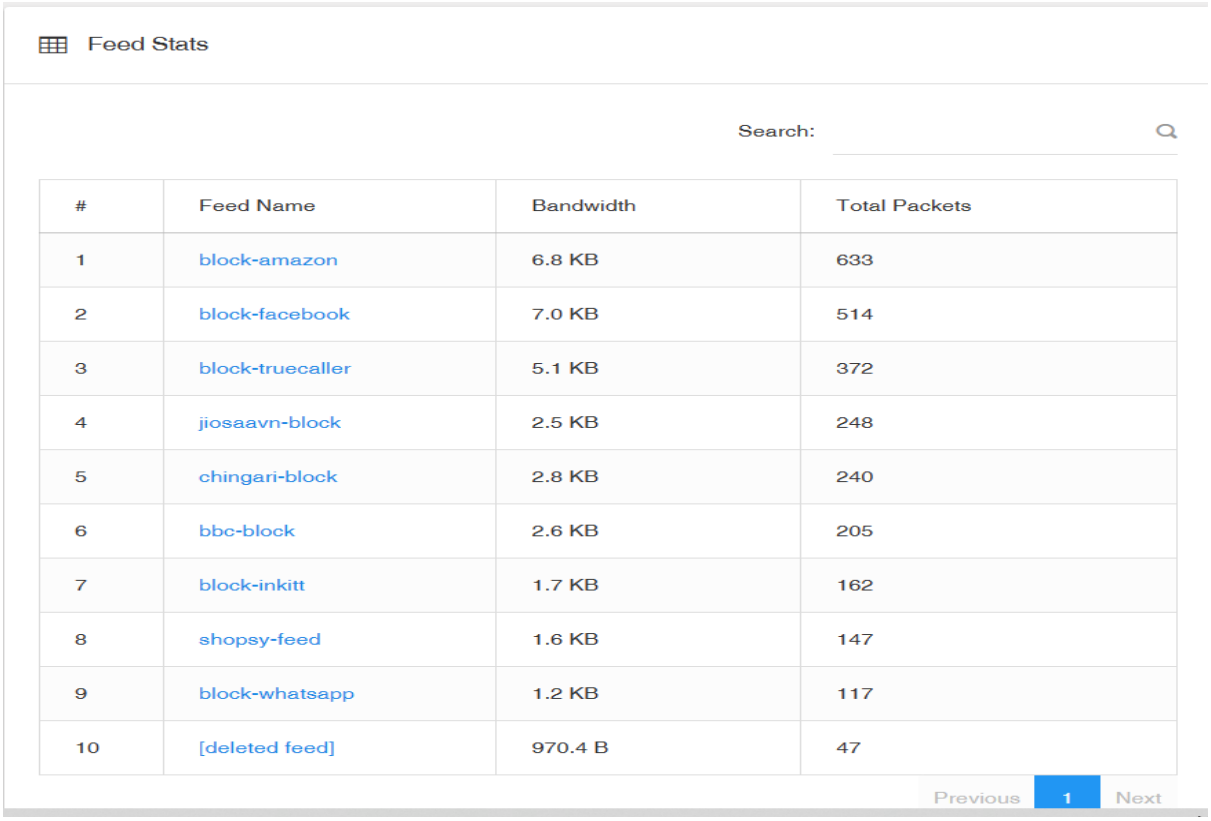


Figure 6-19

7. Access Control

7.1 To add source (websites or domains) that are to be filtered

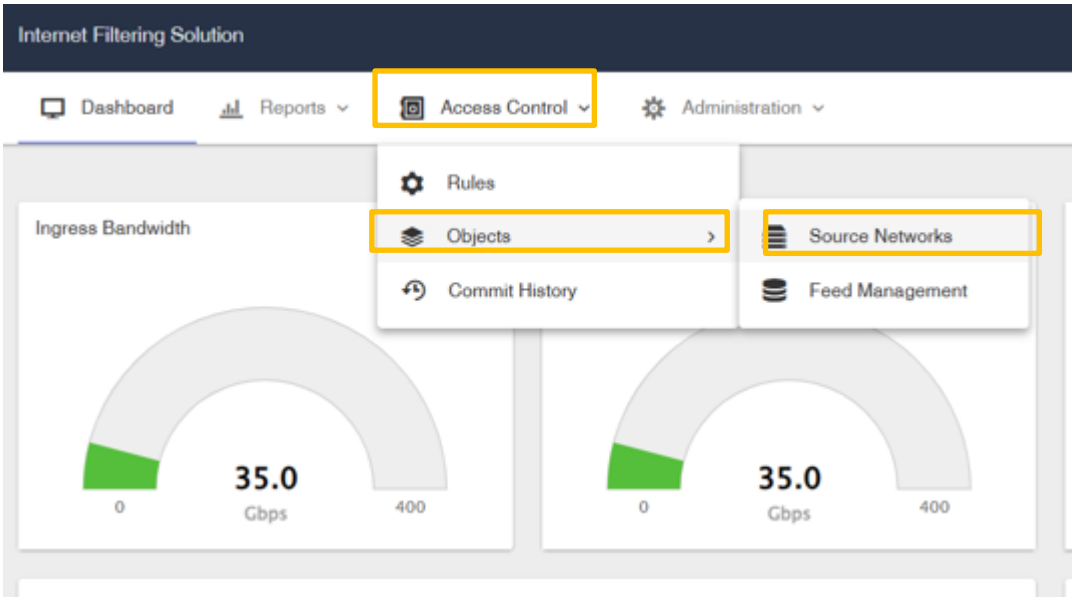


Figure 7-1

a. Click Access Control Menu
b. Click Objects
c. Click Feed Management
d. On clicking Feed Management, the user is redirected to the Create Feed page.



Figure 7-2

The user can select the type of feed from amongst the following types:

a. IPv4

b. Port

c. FQDN

d. Application

e. IPv6

f. IPv4 prefix

g. IPv6 prefix

h. Wildcard domain

There are two options for providing the list of sources under the given feed:

Upload .txt file or .csv file that contains the list of sources to be filtered.

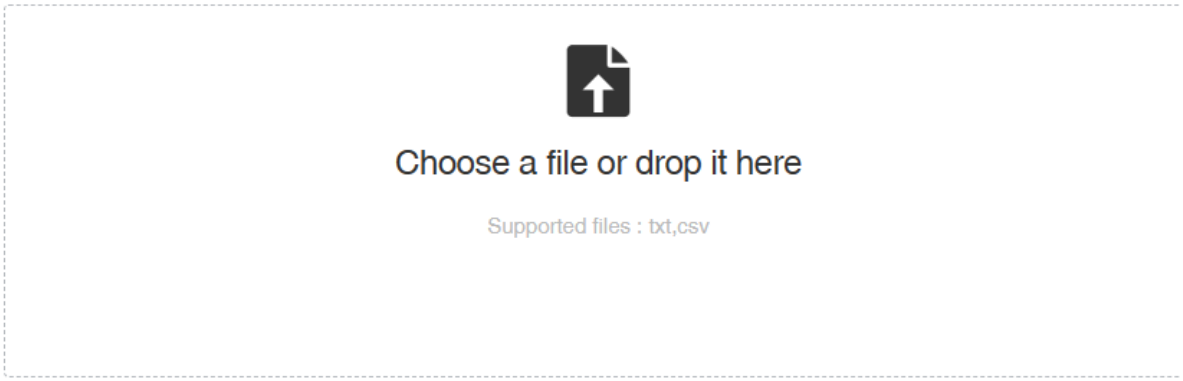


Figure 7-3

The user can individually enter the values as per the type selected and separated by comma.



Figure 4-4

Once the sources are provided, the user can click the 

button to finish creating the feed.

7.2 Feed Details

The user can obtain details of all the feeds created.

The user can use Search to locate the specific feed for their requirement.

The feed details contain the following headers:

a. Feed name

b. Description

c. Type

d. Time of creation

e. Time of modification

f. Instance

g. Actions that can be performed based on the account privilege.

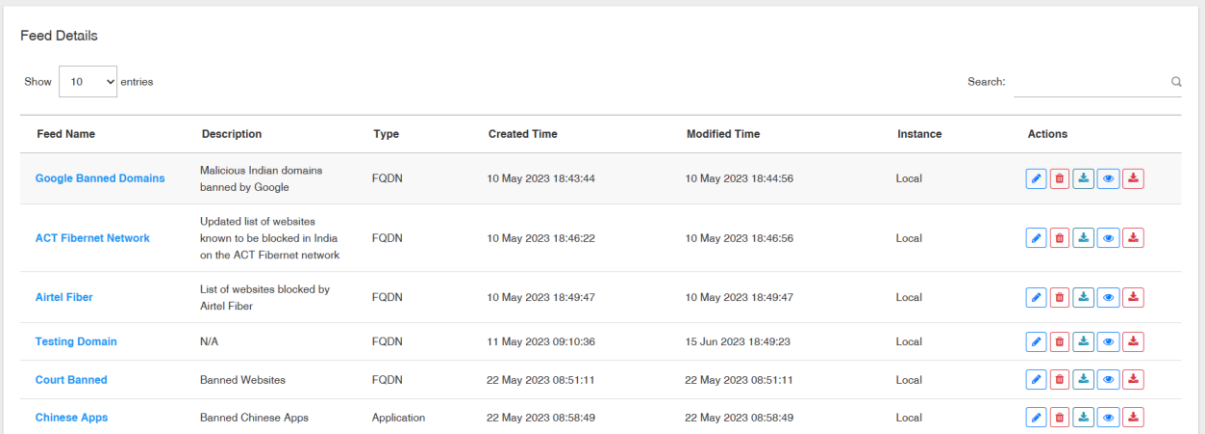


Figure 4-5

h. Actions that can be performed on feed



The user can edit the feed details by clicking this button.

On clicking this button, the Edit Feed Details page is presented on the screen.

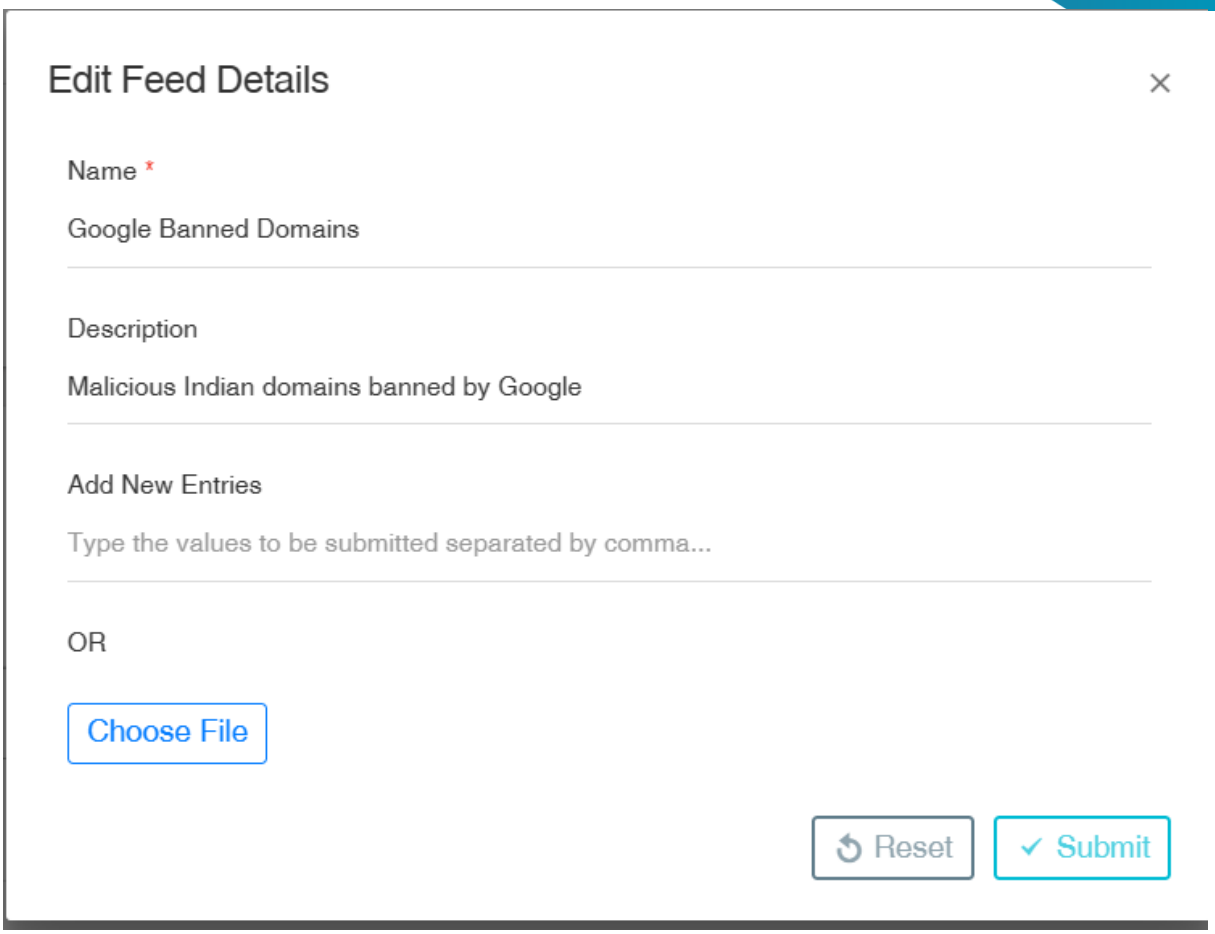


Figure 7-6

The user can perform the required modifications as instructed in the page.

If the user wants to undo the changes, they can click the Reset button.

To accept the changes made, they can click the Submit button.

The user can delete a corresponding feed by clicking the delete button

The user can download the .txt or .csv file containing the source details by clicking the download button.

The user can click the View Rule button to see the rules mapped to the corresponding feed

The user can view any incorrect inputs that are given to the feed by clicking this button.

To Add source device for whom the filtering rules must apply

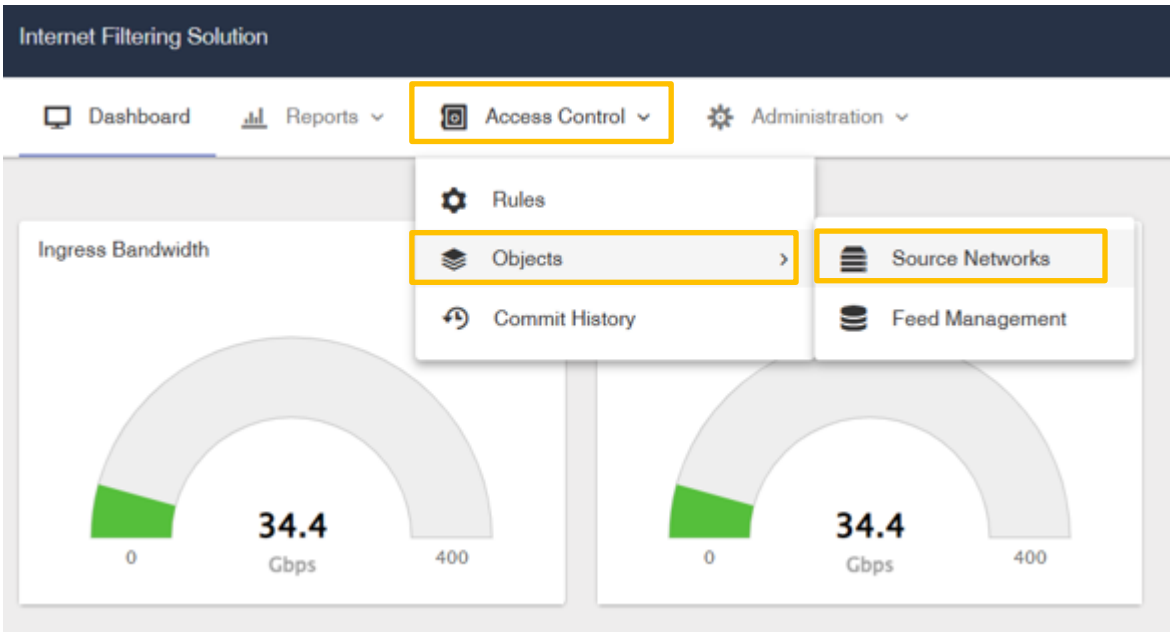


Figure 7-7

a. Click Access Control menu

b. Click Objects

c. Click Source Networks.

The user is then redirected to the Source Details page

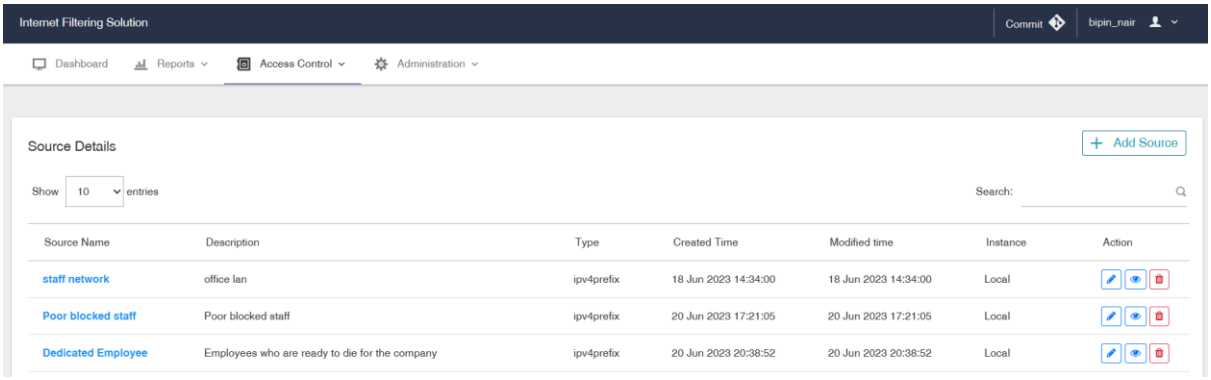


Figure 7-8

The user can view the list of sources created in this page.

The user can search for a specific source by entering the keyword in the Search: field.

The user can edit the corresponding source by clicking this button.

 The user can view the corresponding source by clicking this button.

 The user can delete the corresponding source by clicking this button.

To add a new source, the user can click  at the top right of the page.

When the user clicks the  button, the Source Upload page is presented on the screen.

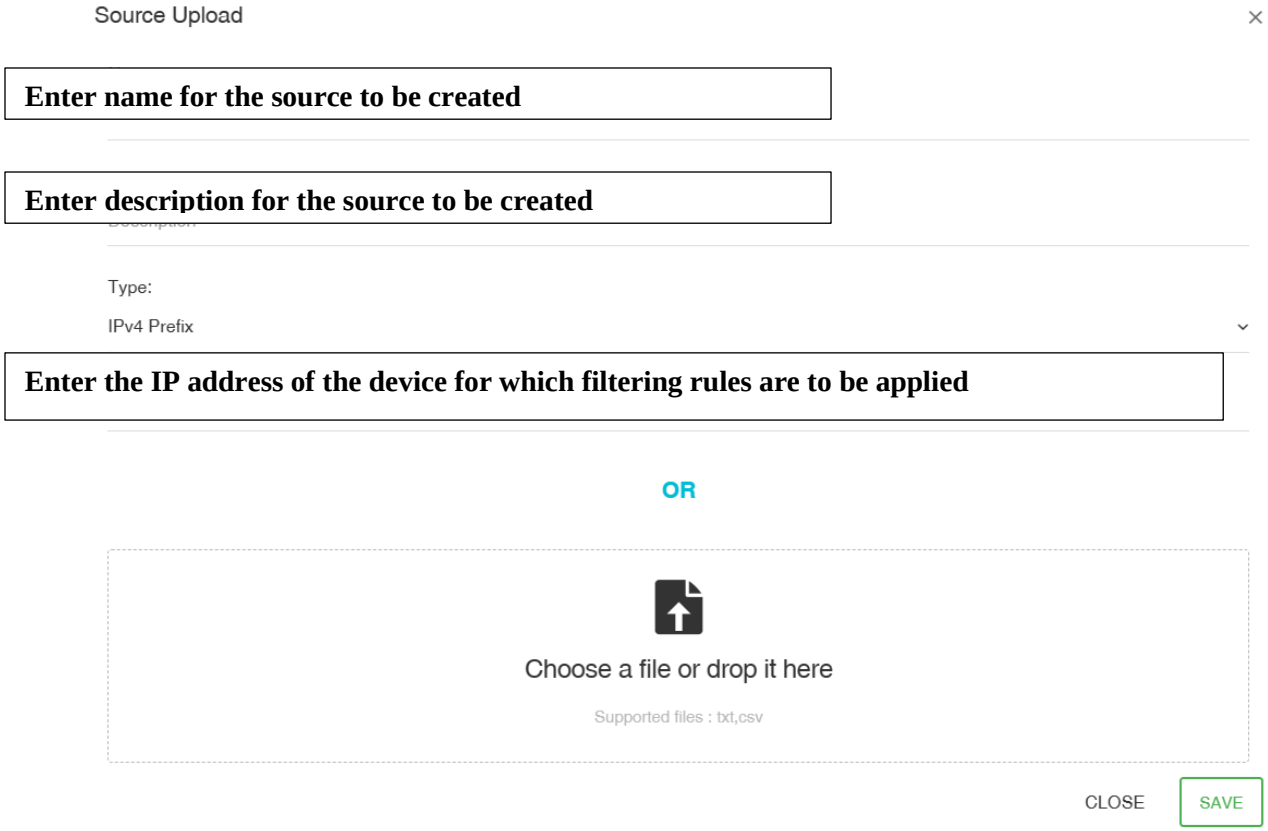


Figure 7-9

Note: While creating Source, under Type, the user can select from IPv4 prefix or IPv6 prefix.

Note: The user can also enter consolidated list of IP addresses (if the rules are to be applied for multiple users) either in the form of .txt file or .csv file.

The user can click  button to save the source.

7.3 For creating filter rules

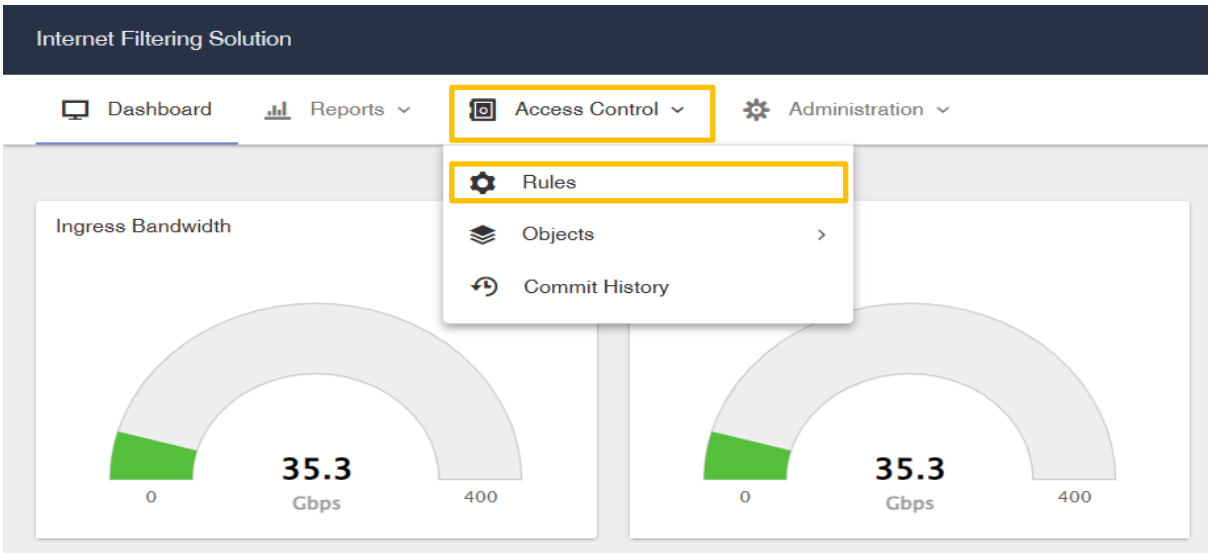


Figure 7-10

a. Click Access Control menu

b. Click Rules

Once the feed and source are created, the user can then create rules. There are two types of rules:

a. Blacklist

b. Whitelist

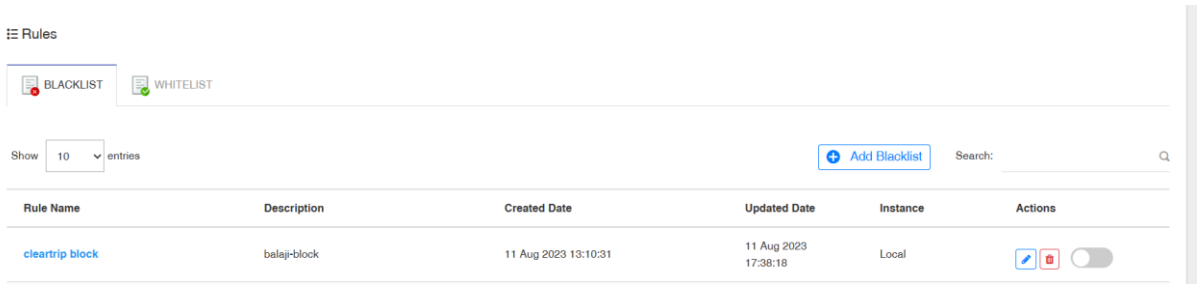


Figure 7-11

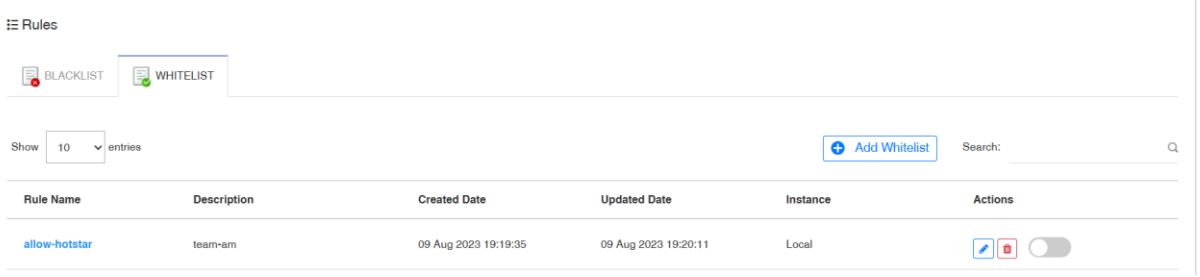


Figure 7-12

Blacklist filtering blocks the traffic from the list of domains provided under the feed and allows the remaining traffic to the source devices for which the rules are applied.

Whitelist filtering allows only the traffic from the list of domains provided under the feed and blocks the remaining traffic to the source devices for which the rules are applied.

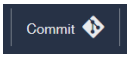
The user can view the list of blacklist rules and whitelist rules in the rules page.

The user can search for a specific filter rule using the search field 

 The user can edit the corresponding rule by clicking this button

 The user can delete the corresponding rule by clicking this button

 The toggle button can be used to enable or disable the rule. Either of the action is referred as Commit.

The user can commit a filter rule by clicking the 

button above the menu bar.

7.4 Creating blacklist rule

For creating a new blacklist rule, the user can click the 

button.

On clicking the Add Blacklist button, the Add Blacklist page is presented on the screen.

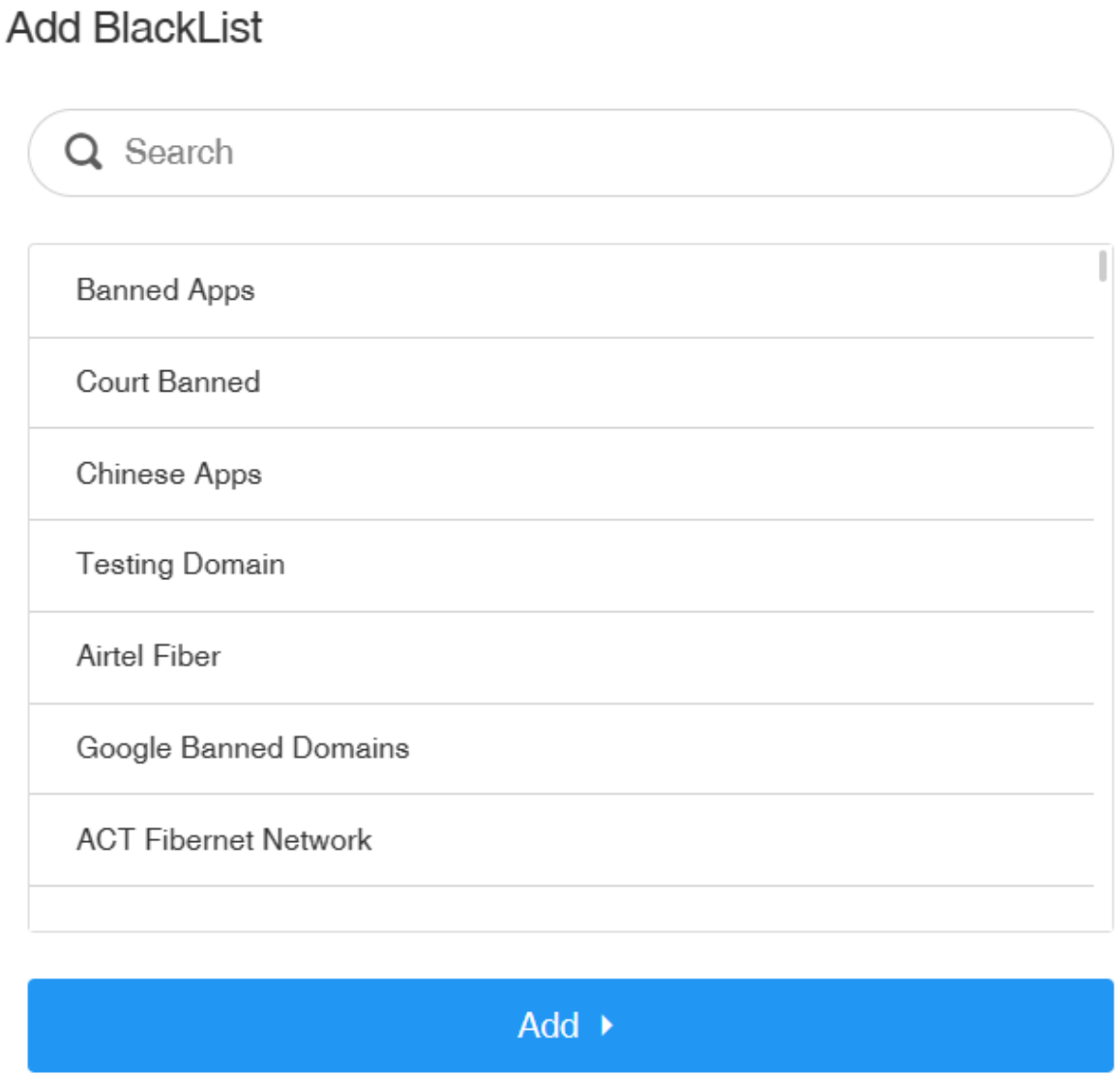


Figure 7-13

The user can add the Feed (list of domains) in the blacklist filter by clicking on the corresponding feed name and clicking add. The user can add multiple feeds to blacklist filter. The user can find their required feed name by using the search field.

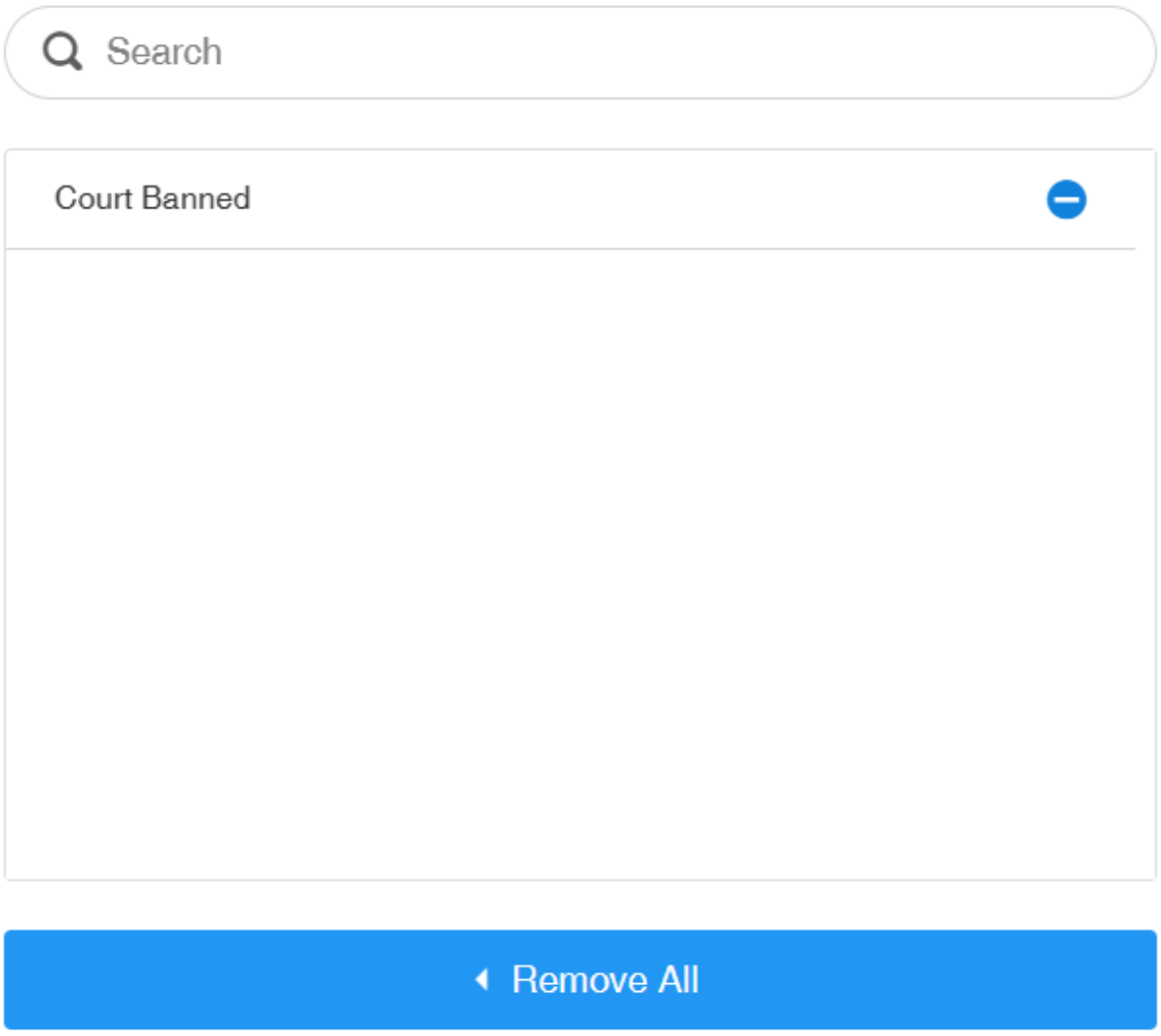


Figure 7-14

The user can view the feeds added to the proposed blacklist filter on the right side of the page. The user can remove any specific feed by clicking the button.

If the user clicks the Remove All button, then all the feeds from the filter will be removed.

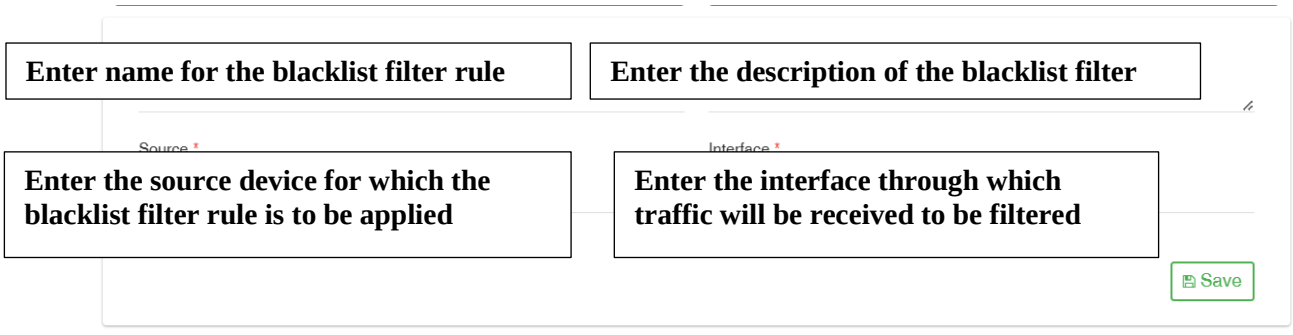


Figure 7-15

The user can save the blacklist filter rule by clicking the Save button.

7.5 Creating whitelist rule

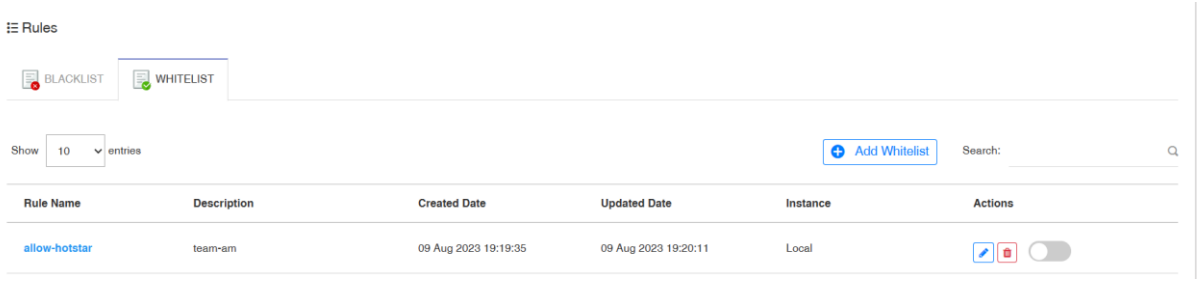


Figure 7-16

To create a new whitelist rule, the user can follow the same process that was discussed for creating blacklist rule after clicking the Add Whitelist button.

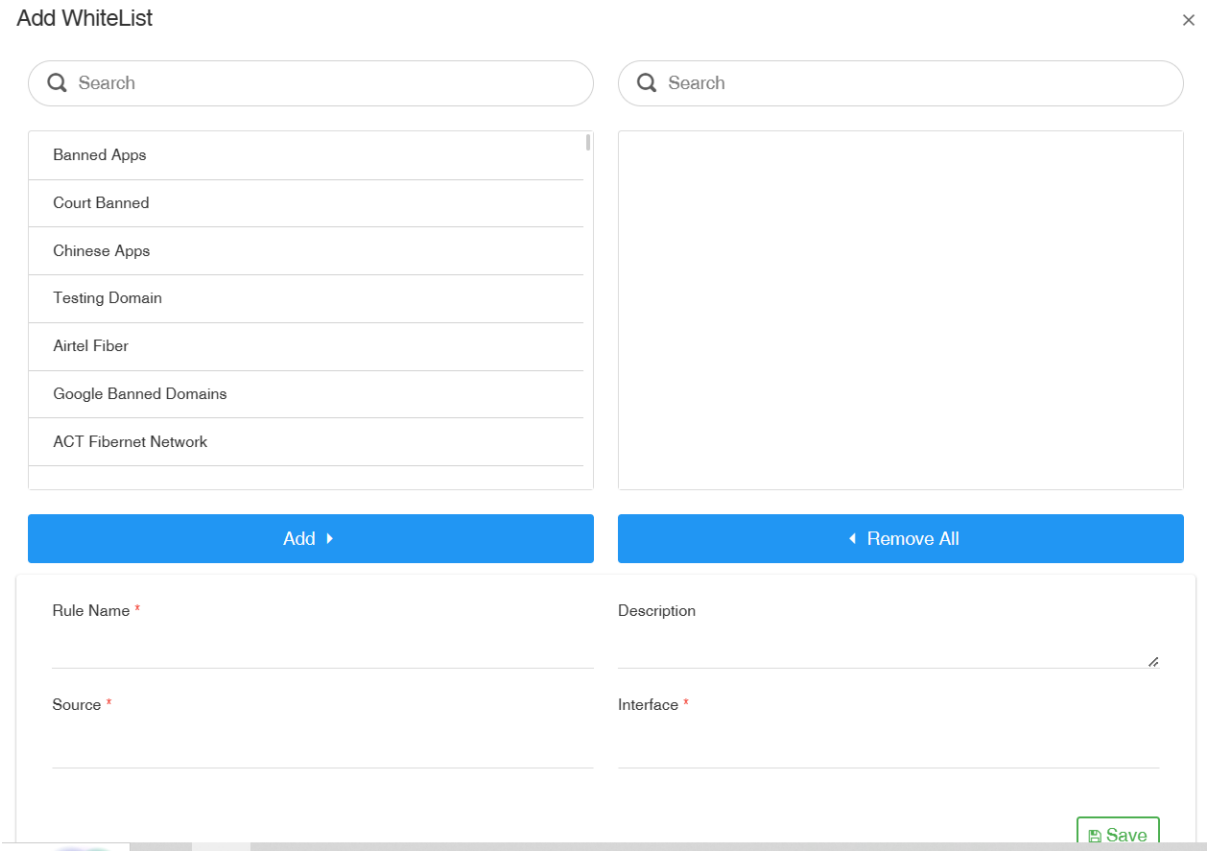


Figure 7-17

7.6 Executing filter rule

The user can execute a blacklist or whitelist rule by clicking the Commit button available at the top of the menu bar.

The user must note that even though they click the toggle button to enable or disable the blacklist/whitelist filters, however, these rules are not executed unless the commit button is clicked.

When the commit button is clicked, the user will be prompted to enter the description or commit message that can act as reference for the user.



Figure 7-18

If the user wants to view the history of commits made, they can click List Commit History button.

Alternatively, the user can also obtain commit history by

Clicking Access control menu

Then click Commit History

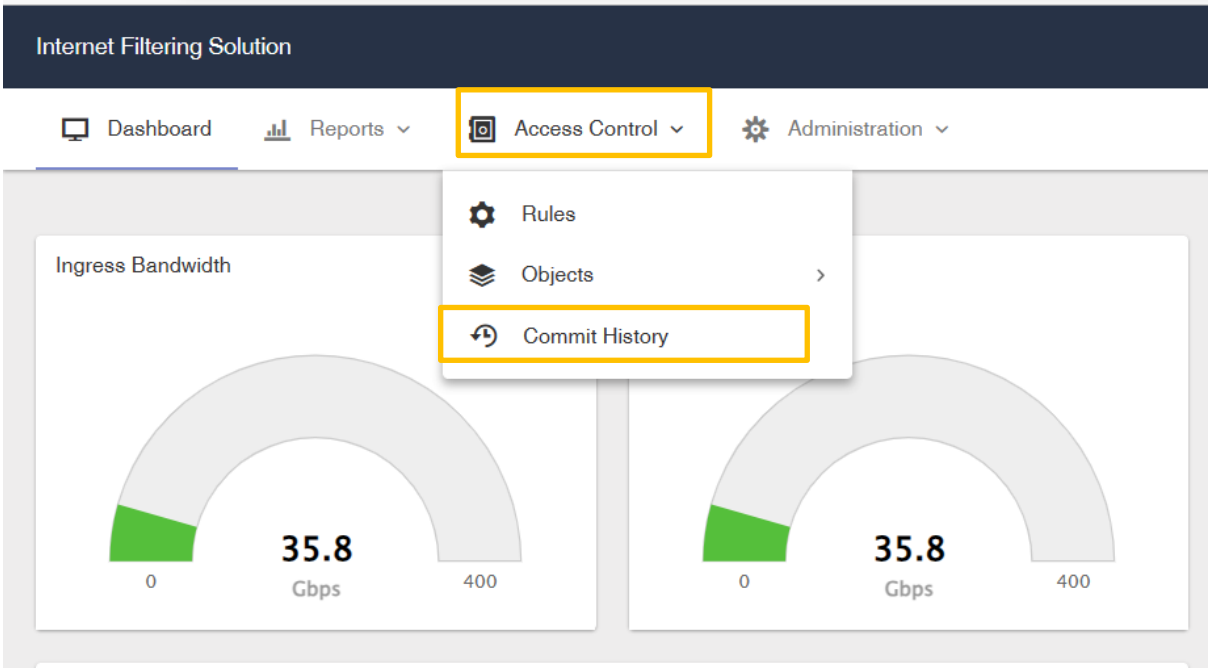


Figure 7-19

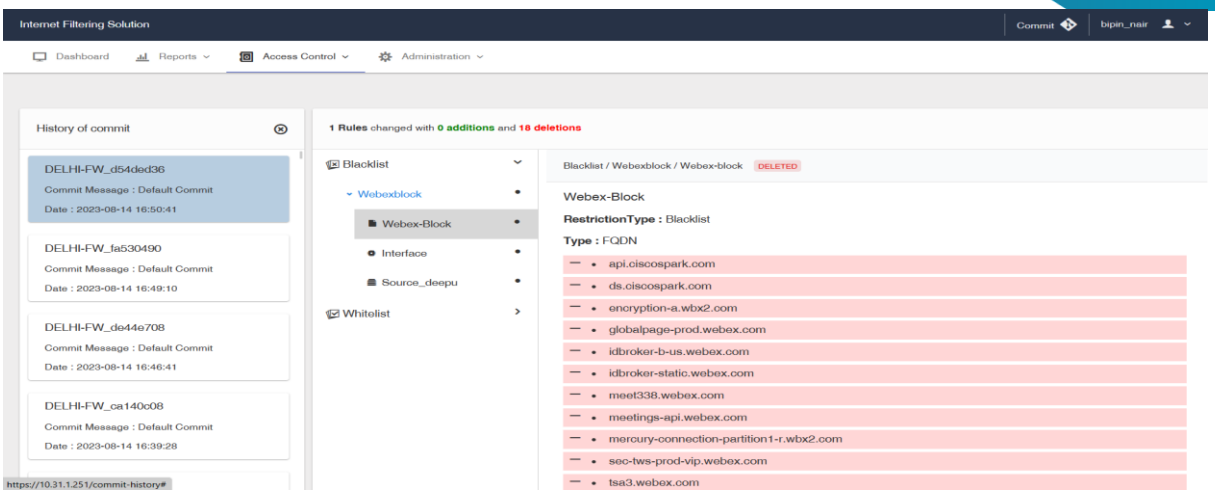


Figure 7-20

The user can view the additions and deletions made, the date and time of the modification along with description of the changes made for both blacklist rules and whitelist rules.

8 Administration

a. To create a new user

b. Click Administration menu

c. Click User Management

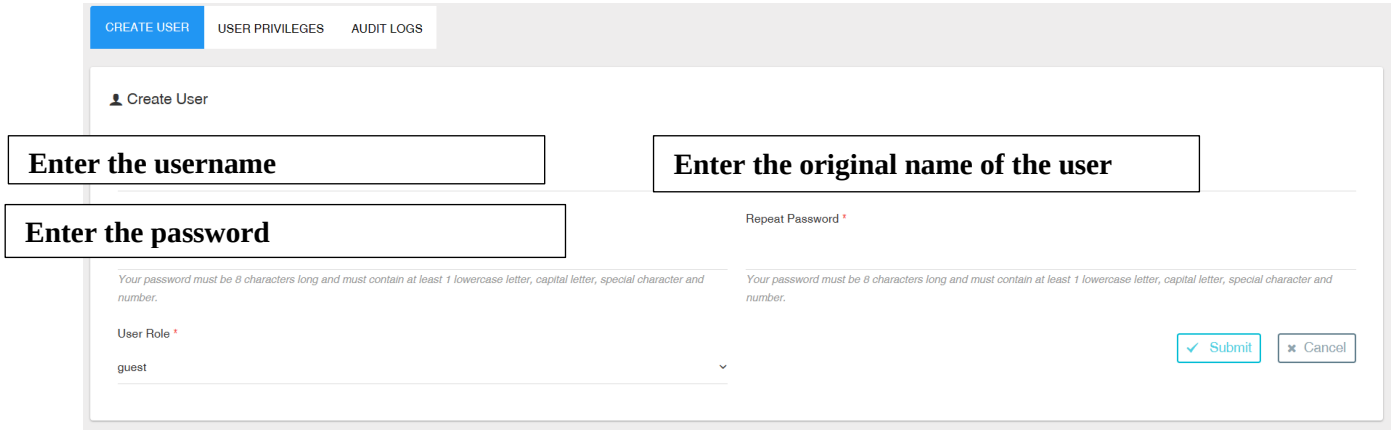


Figure 8-1

The user roles can be selected from amongst the list.

To execute the creation of a new user, click Submit button.

To cancel the creation of a new user, click Cancel button.

The list of existing users can be found in the User management widget at the bottom of the page

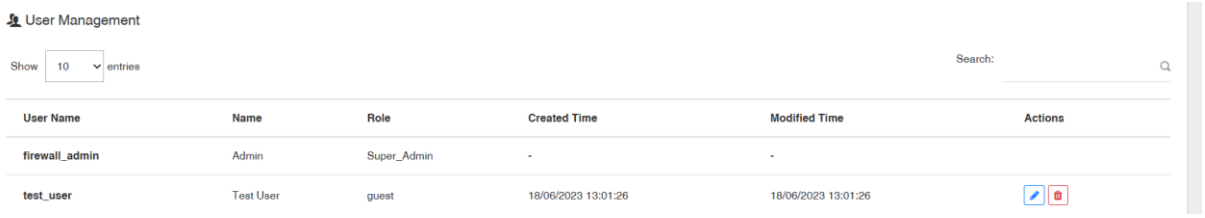


Figure 8-2

To edit the corresponding user details

To delete the corresponding user

8.1 To create user privileges

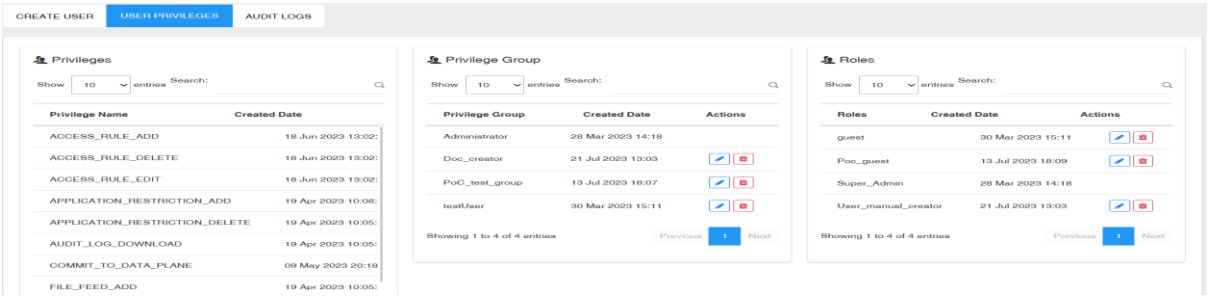


Figure 8-3

The list of permissible privileges can be found under the Privileges list on the left part of the page.

The privileges are grouped to form privilege group.

The privilege group is mapped to a Role as per the discretion of the super administrator or as per the organization policy.

 To edit corresponding privilege groups or roles

 To delete corresponding privilege groups or roles.

Note: Privilege can only be modified by super administrator. Thus, certain functionalities under administration menu might be restricted for those users whose privilege are below the super administrator.

Note: The role of super administrator cannot be modified or deleted by those users whose privilege is below the super administrator.

All actions performed by the users can be viewed under User Logs. Thus, the user behavior can be monitored and tracked using the logs.

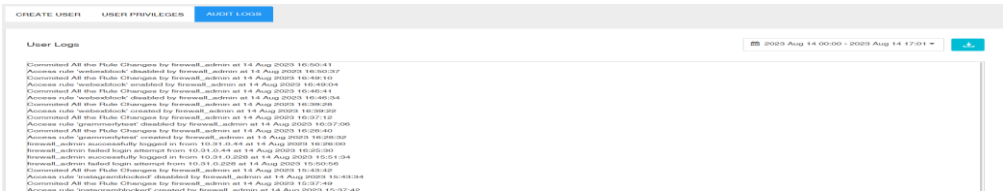


Figure 8-4

The logs can be filtered based on specific date and time.

 To download the user logs.

9 Applications

Application signatures are used along with filtering rules to manage the traffic from the corresponding application.

9.1 To obtain the list of application signatures available in CISAI Internet Filtering Solution

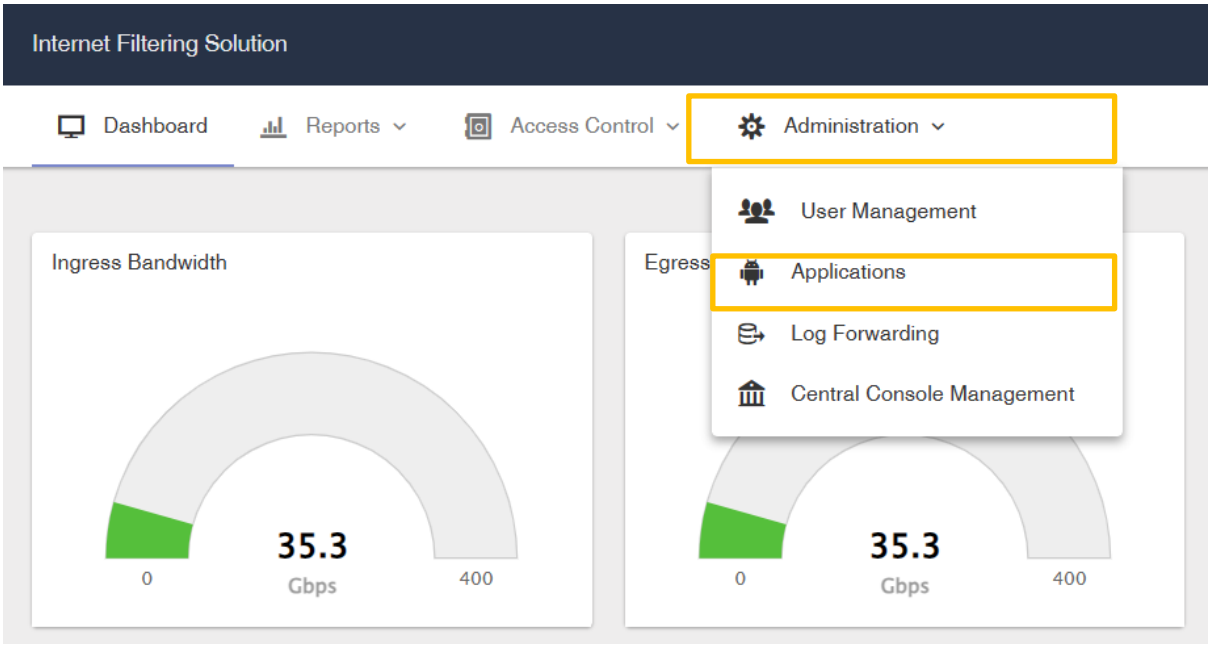


Figure 9-1

a. Click Administration menu

b. Click Applications

c. The user will be redirected to the Applications page

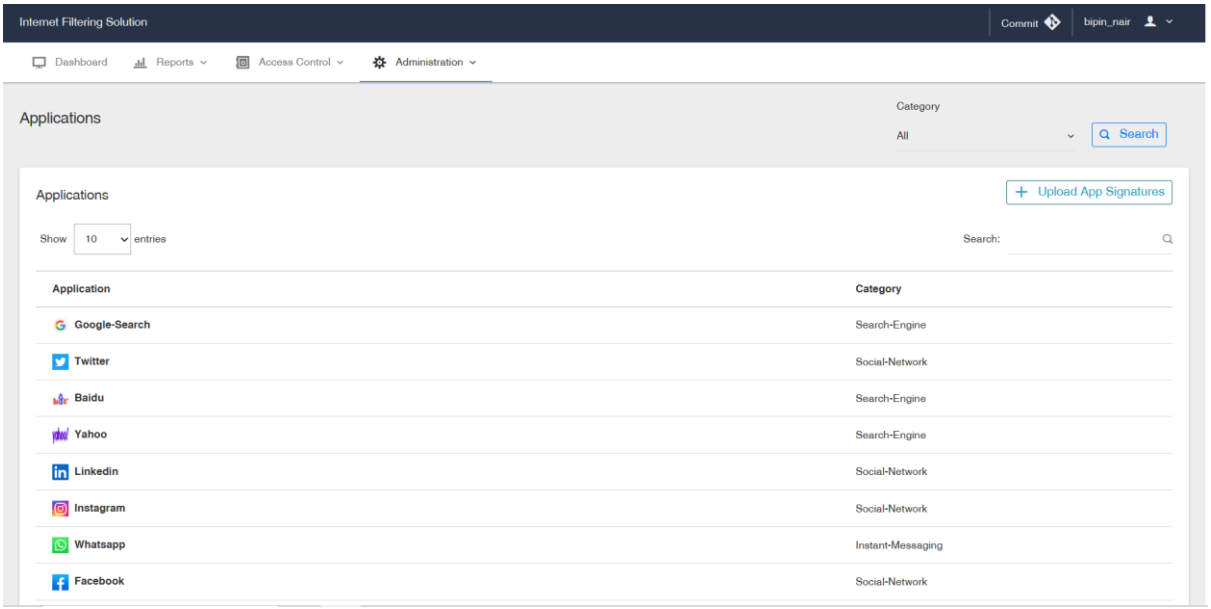


Figure 9-2

The user can filter the application signatures based on application type using the filed provided at the top-right portion of the page.

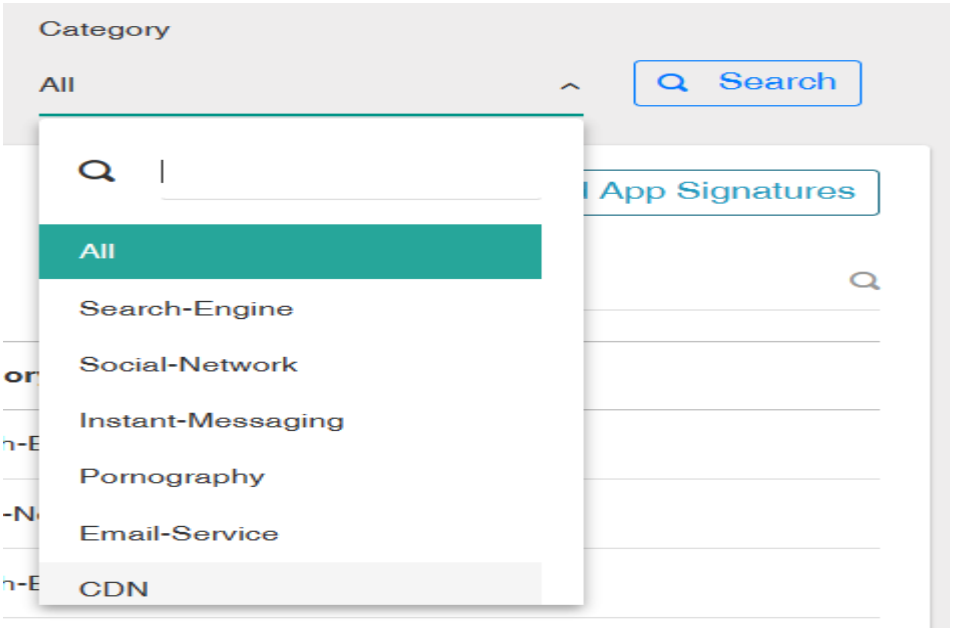
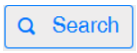


Figure 9-3

The application types can be obtained by clicking the down arrow at the top right of the page to the left of 

button.

Once the filtered list of application signatures is obtained, the specific application filter can be located using the 

field.

The list consists of two columns with application name in first column and the application type in the second column.

Application	Category
Google-Search	Search-Engine
Twitter	Social-Network
Baidu	Search-Engine
Yahoo	Search-Engine
Linkedin	Social-Network
Instagram	Social-Network
Whatsapp	Instant-Messaging
Facebook	Social-Network
Bing	Search-Engine
Telegram	Instant-Messaging

Figure 9-4



Is used to edit the number of entries in the search result. By default, ten entries are shown per page in the search list.

9.2 To upload custom app signature

Click 

on the right side of the application page.

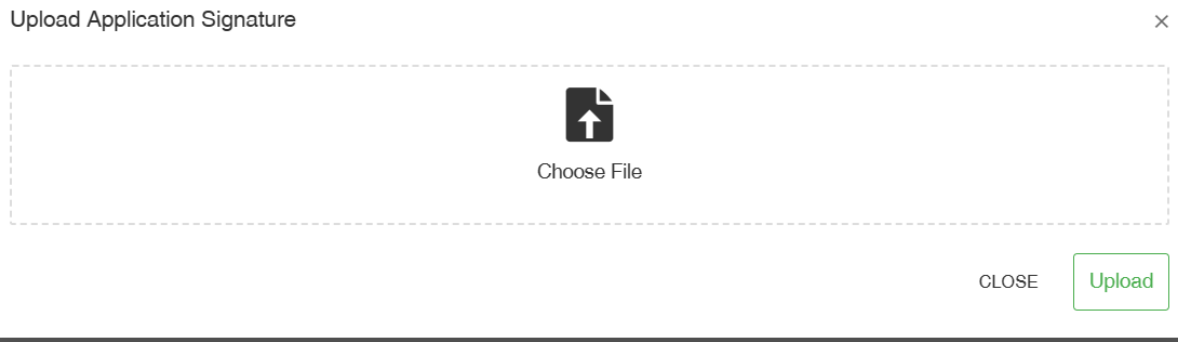


Figure 9-5

a. Click Choose File to select the application signature
b. Click upload

Note: To exit from the upload box, click Cancel

10 Log forwarding

Log forwarding collects the logs and forwards them to a centralized log storage device like OPLC or directly to SIEM.

In CISAI Internet Filtering Solution, the logs are forwarded using syslog protocol.

The syslog server can be configured only by the super administrator or any user who has the corresponding privilege granted by the super administrator.

10.1 For Enabling/Disabling log forwarding

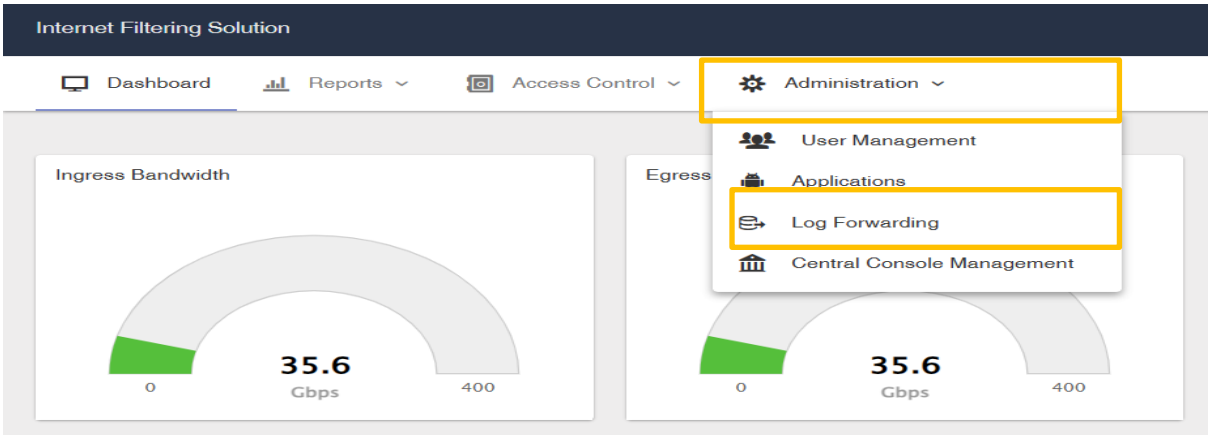


Figure 10-1

a. Click Administration menu

b. Click Log Forwarding

The user will then be redirected to the Log Forwarder page

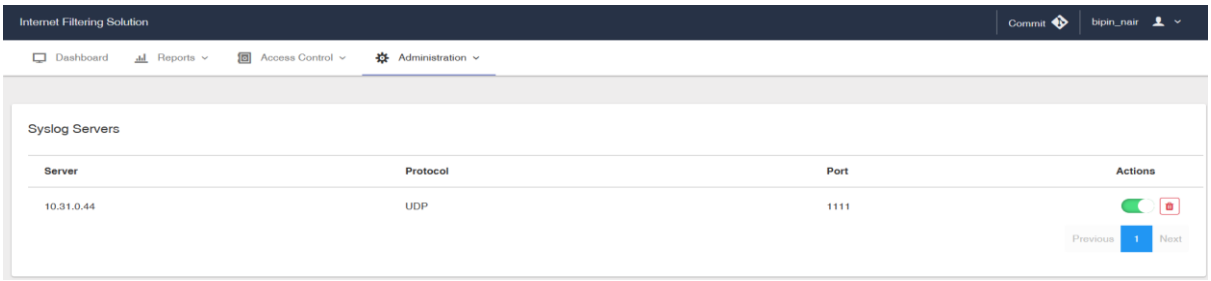


Figure 10-2

Click button to enable or disable log forwarding.

Can be used to delete the syslog servers added for log forwarding.

Note: New syslog servers can only be added by super administrator or any user granter similar privilege by the super administrator.

11 Central Console Management

A centralized management system can consolidate all the internet filtering into one workstation and under the control of a single set of processes thereby reducing the time and effort. It also saves moving from one console to another and having to remember disparate settings, configurations, and deployment details.

11.1 To access central console

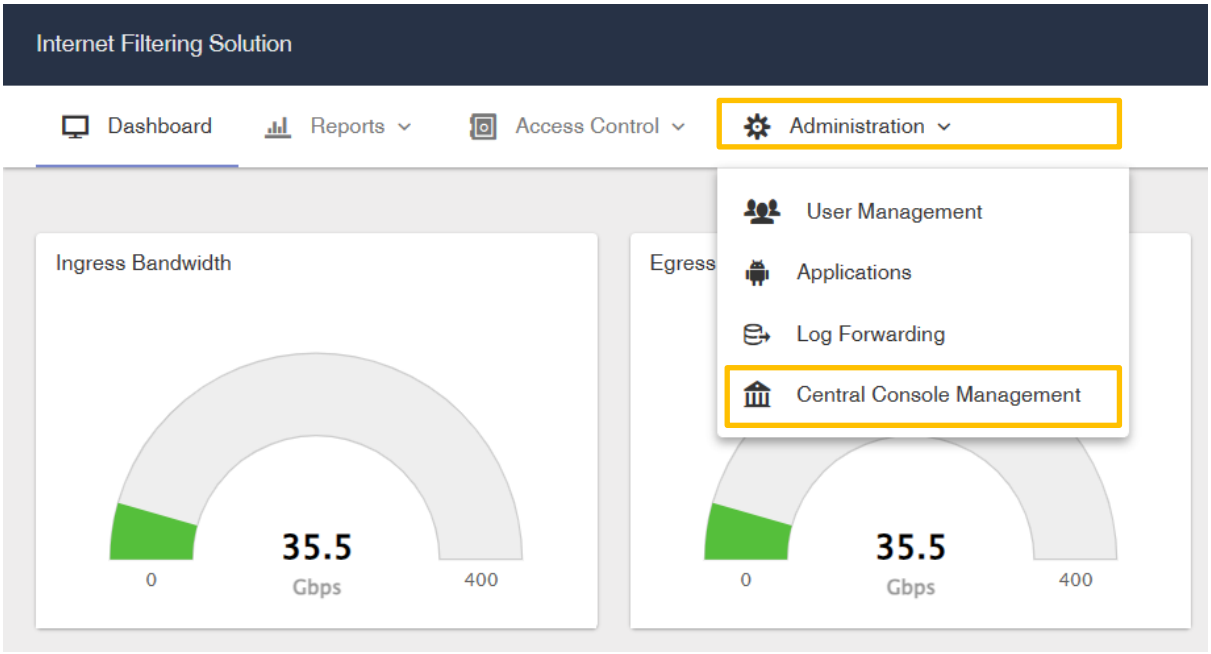


Figure 11-1

a. Click Administration menu

b. Click Central Console Management

The user is then redirected to the Central Console page

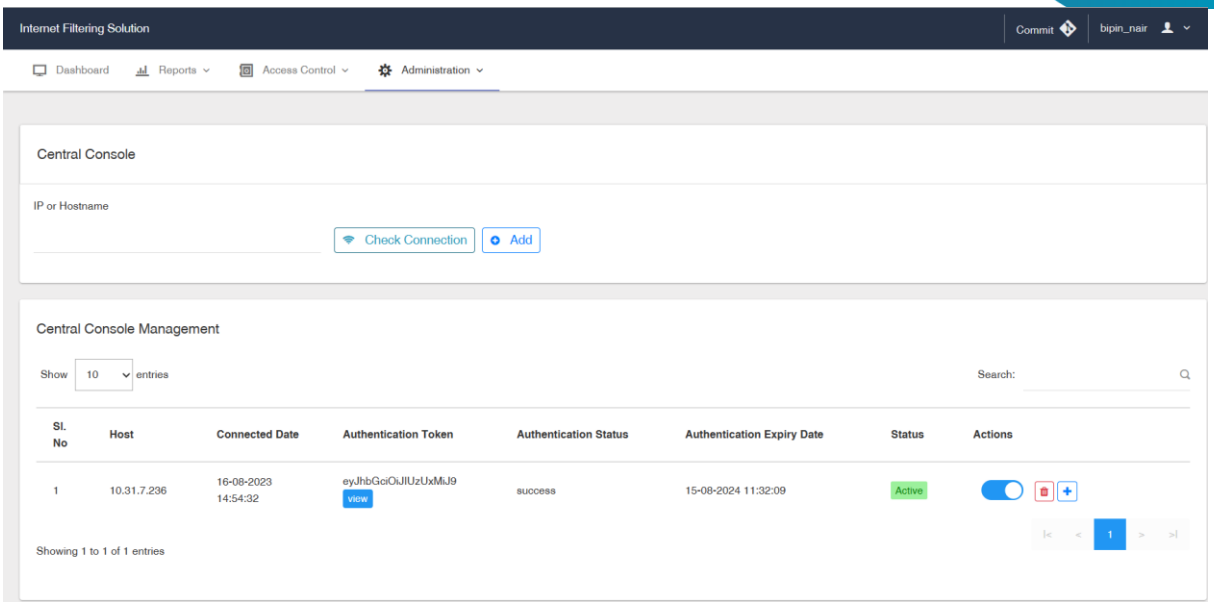


Figure 11-2

11.2 To add central console

a. Enter the IP address of the central console in the field provided on the top left

b. To check the status of the device, click 

button.

c. To add the central console, click 

button.

d. The details of the central console can be seen in the Central Console Management widget

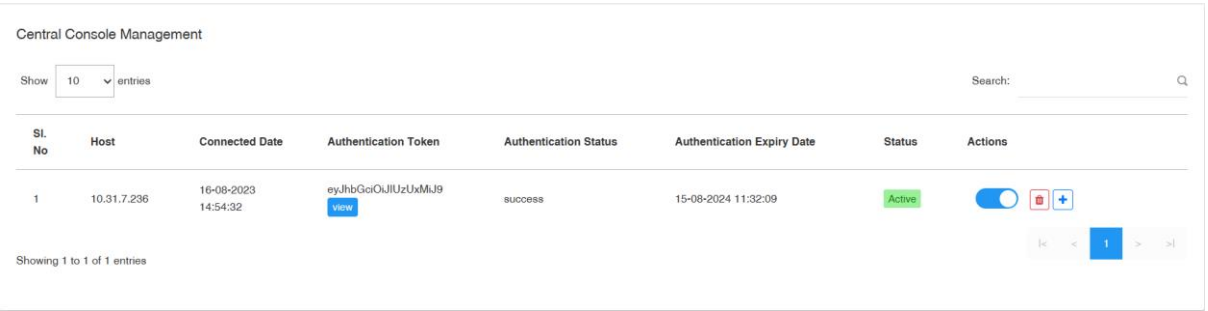


Figure 11-3

 This toggle button can be clicked to enable or disable the central console management.

 This can be used to delete the central console.

 This can be used to add another central console to the internet filtering solution.